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Spring phytoplankton bloom dynamics reveal thresholds of ocean alkalinity enhancement in the North Sea



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Spring phytoplankton blooms drive primary production and food web dynamics in marine ecosystems. Ocean alkalinity enhancement, a proposed carbon dioxide removal strategy, alters seawater chemistry and could disrupt planktonic food webs, yet its ecological consequences remain poorly understood. We conducted a 39-day mesocosm experiment in spring 2023 in the German North Sea to assess the effects of calcium-based alkalinity additions on phytoplankton dynamics. Twelve 6-m³ mesocosms contrasted immediate dilution (well-mixed conditions), and delayed dilution (poorly mixed conditions) across an alkalinity addition gradient of ~0–1150 $\mu\text{mol kg}^{-1}$. We show that phytoplankton bloom and particulate organic matter dynamics are largely unaffected at low to moderate alkalinity additions of ~0–700 $\mu\text{mol kg}^{-1}$ regardless of dilution scenario. At alkalinity additions $\geq 900 \mu\text{mol kg}^{-1}$, bloom development is delayed by ~6–9 days and particulate organic phosphorus is significantly reduced in both dilution treatments, while particulate organic carbon and nitrogen are lower only under delayed dilution. These findings identify critical alkalinity thresholds for phytoplankton bloom development in temperate, diatom-dominated spring blooms, informing boundaries for ecologically safe deployment.

Meeting the Paris Agreement targets of limiting global warming to well below 2 °C requires not only deep emission cuts but also large-scale carbon dioxide (CO₂) removal (CDR)^{1–3}. Among emerging negative emission strategies, marine CDR (mCDR) offers a unique opportunity to leverage the ocean's vast capacity for carbon storage. One of the most promising mCDR approaches is Ocean Alkalinity Enhancement (OAE), a method that mimics natural chemical weathering by adding alkaline material to seawater, thereby increasing ocean alkalinity and enhancing CO₂ uptake. By shifting the carbonate system toward higher concentrations of bicarbonate (HCO₃[−]) and carbonate (CO₃^{2−}) ions, OAE lowers the partial pressure of CO₂ (pCO₂) in the surface ocean, promoting the influx of CO₂ from the atmosphere^{4–8}. While the geochemical basis of OAE is well understood, its ecological implications remain poorly constrained, particularly under different deployment scenarios.

In previous model assessments, OAE scenarios are typically introduced as a continuous addition of alkalinity over large ocean areas, which would represent a small perturbation at a time on a regional scale^{5,9,10}. Furthermore,

in these model simulations, the added alkalinity becomes vertically distributed (instantly mixed) within the upper mixed layers of coarse resolution (layer depths >20 m). Additionally, the OAE scenarios considered in model applications typically do not account for pulsed or regional additions of alkalinity, as would occur in real-life short-term, point-source deployments. Yet such regional and possibly also pulsed OAE scenarios appear more realistic than continuous large-scale additions. While the modelling exercises are valuable for assessing the theoretical carbon removal potential of OAE applications, they lack realism with respect to real-world deployment conditions because they often cannot resolve the transient, spatially heterogeneous exposure conditions that occur during alkalinity addition, dilution and mixing.

Recent experimental studies that addressed OAE impacts on plankton ecology simulated CO₂-equilibrated conditions where shifts in the carbonate system and changes in pH were minimized by introducing seawater of enhanced alkalinity with dissolved inorganic carbon (DIC) already added^{11–15}. The amount of DIC addition ensured that the alkalised seawater's

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CO₂ concentrations closely matched ambient surface waters. Despite minor ecological impacts observed so far under these conditions^{13,16,17}, most CO₂-equilibrated OAE applications require expensive infrastructure and are energy intensive¹⁸, limiting their applicability. Lower energy and relatively cheaper approaches may be composed of non-CO₂-equilibrated OAE scenarios, whereby finely ground minerals such as brucite and olivine, lime-based slurries such as Ca(OH)₂ or alkaline electrochemical products are added to the surface ocean^{18,19}. Under non-CO₂-equilibrated OAE, the alkalized seawater should eventually re-equilibrate over time through air-sea gas exchange²⁰. However, due to the physical constraints on gas exchange, equilibration is time dependent^{21,22}. As a result, this approach can expose marine communities to transient but strong shifts in carbonate chemistry, such as increased pH (reduced H⁺) and carbonate saturation states, and low pCO₂, before the alkalized water fully equilibrates or gets diluted. Thus, while non-CO₂-equilibrated OAE approaches represent a more feasible operational approach in terms of deployment, scale, logistics, energy requirements and costs, they pose challenges due to the strong carbonate system changes that could potentially stress marine planktonic communities.

In scenarios of high alkalinity doses in non-CO₂-equilibrated OAE applications, the decline in pCO₂ can be sufficient to induce CO₂ limitation for phytoplankton, causing slowed phytoplankton growth²³. Even though carbon concentrating mechanisms (CCMs) of phytoplankton enable carbon fixation by utilizing HCO₃⁻^{24,25}, in most cases they cannot fully compensate for strong CO₂ limitation, as HCO₃⁻ must be converted to CO₂ for carbon fixation. A steep CO₂ gradient between the in- and outside of the phytoplankton cell can lead to CO₂ leakage before being incorporated into biomass²⁶. Consequently, strong declines in pCO₂ may not only suppress overall phytoplankton growth but also favour phytoplankton species with highly efficient CCMs such as diatoms^{25,27,28}, potentially shifting phytoplankton community composition under intense OAE application. Beyond direct carbonate chemistry effects, elevated pH under OAE conditions may also indirectly affect phytoplankton through reduced bioavailability of essential trace elements such as iron^{29,30}, and through shifts in nitrogen speciation that favour unionized ammonia over NH₄⁺³¹, which can be toxic to phytoplankton at elevated concentrations³².

Furthermore, the elevated pH associated with non-CO₂-equilibrated OAE applications is a critical factor influencing various biogeochemical processes, including heterotrophic microbial activities^{20,33}. Altered heterotrophic processes may affect the remineralization of organic matter, with potential consequences for organic matter stoichiometry, carbon storage and nutrient regeneration in the perturbed system^{34–36}. Microbial responses to pH shifts have been well studied in the context of ocean acidification, where decreasing pH typically suppresses nitrification rates³¹, bacterial production, species richness, evenness and overall diversity³⁷ and enhances polysaccharide degradation³⁸. However, the effects of ocean acidification on enzyme activity vary depending on enzyme type and location³⁹. In contrast, much less is known about microbial dynamics under OAE, where the carbonate system shifts in the opposite direction, toward higher pH levels.

Ecological and biogeochemical responses to non-CO₂-equilibrated OAE are expected to be intricately linked to the mode of deployment, the hydrographic conditions and ecosystem state¹⁸. For instance, ship-based dispersal of alkaline material would occur while the vessel is in motion, leading to rapid lateral dilution as the material is spread over a broad surface area^{40,41}. This results in relatively short-lived and less intense carbonate system perturbations depending on the alkalinity level. In contrast, point-source deployments, such as discharge from a factory outfall or pipeline, are likely to have slower dilution due to limited water movement and reduced mixing⁴². This can generate more localised but intensified and prolonged exposure of the planktonic community to comparatively stronger carbonate system changes, thereby amplifying ecological responses. Depending on the duration of exposure, the ecosystem state, such as during a bloom or post-bloom, and the intensity of alkalinity perturbation, biogeochemical effects may range from transient and reversible to more persistent effects with the potential to cascade through the food web.

To assess the potential ecological impacts of a non-CO₂-equilibrated OAE scenario on a near-natural planktonic community, we conducted a 39-day-long mesocosm experiment in the temperate, eutrophic waters of the German North Sea on Helgoland. A gradient of six alkalinity levels was applied, from low to extreme levels reaching up to a maximum of additional alkalinity of ~1900 μmol kg⁻¹ (measured Δtotal alkalinity (TA) during a short-term 48-h exposure phase simulating delayed dilution of alkalized waters). These levels were chosen to (i) represent strong exposure scenarios that may occur in proximity to potential outfall locations and (ii) allow for the identification of potential threshold responses in the phytoplankton community. Previous mesocosm experiments investigating the environmental safety of OAE have reported minor or no impacts on planktonic communities under oligotrophic conditions^{11,13,15,17}. In contrast, our experiment was designed to assess plankton responses under more eutrophic conditions associated with a phytoplankton spring bloom. The choice of the spring season was therefore motivated by the aim to investigate biogeochemical responses to OAE during an active phase of phytoplankton bloom development, rather than to identify the optimal timing for OAE deployment.

Our results reveal a critical OAE threshold at alkalinity additions of ~700–900 μmol kg⁻¹. Below this threshold, the ecosystem remained largely unaffected, whereas above it, significant delays in phytoplankton bloom development and lower concentrations of specific particulate organic matter components were observed. Furthermore, while the phytoplankton community showed recovery potential under delayed dilution once alkalinity was reduced below the threshold, its responses across multiple ecological endpoints (e.g., nutrient utilisation and particulate organic carbon and nitrogen concentrations) were comparatively more pronounced than under immediate dilution, even at equivalent post-mixing alkalinity levels, underscoring the importance of minimising prolonged exposure to strong carbonate chemistry conditions.

Results

Temperature, salinity and irradiance

During the course of the experiment, temperature and light conditions inside the mesocosms closely mirrored those in the surrounding waters. Water temperatures increased steadily from 5.5 °C at the start to 8.5 °C by the end of the experiment, reflecting seasonal warming (Supplementary Fig. 1). Across all mesocosms, the profiles of photosynthetically active radiation normalized to the surface (PAR_{norm}) showed consistent light attenuation to depth comparable to the surrounding waters, with the sampled water column receiving >50% of incoming PAR at the surface throughout the experiment (Supplementary Fig. 2). This indicates that light availability was unlikely to be a limiting factor for phytoplankton. The overall similarity in light and temperature conditions between mesocosms and the surrounding waters confirms that the experimental approach realistically captured the natural hydrographic setting.

At the beginning of the experiment (D1), the average salinity across mesocosms was ~31.8 ± 0.0, which is within the expected spring seasonal range for the waters surrounding Helgoland⁴³. Salinity values and their depth profiles were comparable across all mesocosms throughout the experiment, except during the intended layering phase on D5 and D6 in the delayed dilution treatment (Supplementary Fig. 3).

Alkalinity enhancement and carbonate system dynamics

Alkalinity manipulations were conducted on day 4 (D4) to simulate non-CO₂-equilibrated OAE application under two dilution scenarios via the addition of calcium chloride (CaCl₂) and sodium hydroxide (NaOH). In the immediate dilution treatment, six mesocosms were treated with increasing targeted ΔTA in a gradient from 0 to 1250 μmol kg⁻¹ in increments of 250 μmol kg⁻¹ (Fig. 1a). A second set of six mesocosms was used to simulate a delayed dilution of alkalised waters. In the delayed dilution treatment, alkalinity was added only to the upper 0.8 m of each mesocosm, to establish double the target ΔTA only in the top layer (ΔTA = 0, 500, 1000, 1500, 2000, 2500 μmol kg⁻¹; for further details see the “Methods” section).

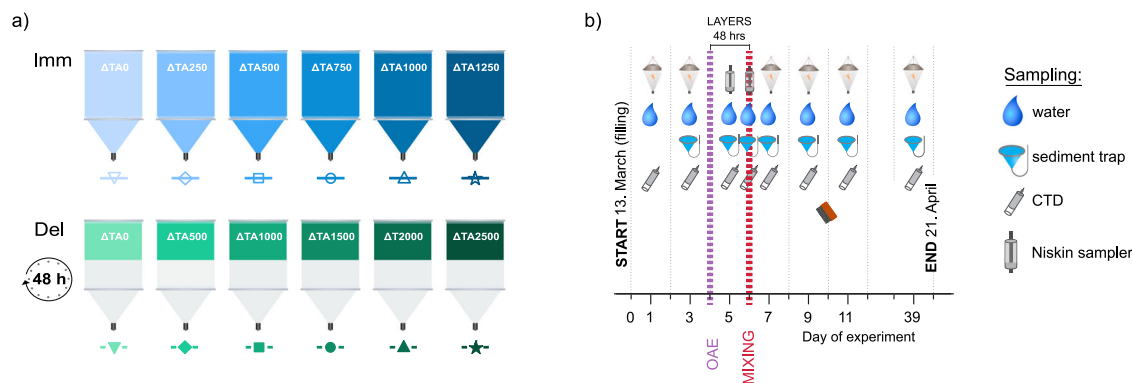


Fig. 1 | Experimental design and timeline. **a** Experimental set-up of alkalinity treatments. Upper row, blue shades: Six mesocosms were treated with alkalinity immediately diluted to target levels (immediate treatment). Lower row, green shades: the other six mesocosms were treated with the same level of alkalinity only in the top 0.8 m and mixed with the untreated bottom layer after 48 h (delayed treatment). The ΔTA levels shown in this figure represent target ΔTA levels. Colour scale corresponds to colours in all plots. Across all figures, immediate dilution treatments

are represented by open symbols and solid lines, while delayed dilution treatments are shown with filled symbols and dashed lines. **b** Timeline of experimental activities, indicating alkalinity addition on D4 (dashed purple line) and water column mixing on D6 (dashed red line). Figure assembled and designed in Inkscape (v.1.4). A detailed timeline over the full experimental period can be found in the Supplementary (Supplementary Fig. 6).

The separation into the two layers was successfully achieved through freshwater addition to the top 0.8 m, as can be seen from the salinity profiles (Supplementary Fig. 4a). pH profiles indicate that the high alkalinity in the top 0.8 m of the delayed dilution treatment persisted for 48 h until mixing (Supplementary Figs. 4b and 5). Nevertheless, across all delayed dilution treatments, a transition zone emerged, in which pH values gradually decreased due to partial mixing between the top layers with enhanced alkalinity and the untreated bottom layers. The vertical extent of that transition zone varied among mesocosms, occasionally extending deeper than initially intended, especially in the two highest ΔTA treatments (Supplementary Fig. 4b).

In the delayed dilution treatment, top layer ΔTA measurements in mesocosms with targeted post-mixing ΔTA levels of 750 and 1250 $\mu\text{mol kg}^{-1}$ (corresponding to targeted top layer ΔTA levels 1500 and 2500 $\mu\text{mol kg}^{-1}$) did not reach their intended levels. The largest deviation from the targeted ΔTA was observed in the highest alkalinity mesocosm (top layer target ΔTA 2500 $\mu\text{mol kg}^{-1}$), where measured top layer ΔTA fell short of the target by 825 $\mu\text{mol kg}^{-1}$ (Fig. 2). These deviations could be attributed to an addition of alkalinity to a larger volume than originally calculated and aimed for and potentially unintended sampling of waters from the transition zone where waters were less alkalinised (>0.8 m), as seen in both salinity and pH profiles (Supplementary Fig. 4). For these two sampling days specifically, CaCO_3 precipitation in analysis bottles from the highest ΔTA level during laboratory analysis also contributed additional uncertainty. Conversely, across all bottom layers in the delayed dilution treatment, which were aimed to be not alkalinised, elevated ΔTA values were measured from discrete samples with ΔTA of up to 1175 $\mu\text{mol kg}^{-1}$, measured in the bottom layer with the highest targeted top layer ΔTA . One possible explanation for this is that the Niskin sampler was not lowered deep enough for the bottom sampling, which resulted in sampling of alkalinised waters from the transition zone between the top and bottom layers (Supplementary Fig. 4b). Target ΔTA concentrations were achieved in the immediate dilution treatment (Fig. 2). As a result, uniform pH changes were measured across depth, following the intended gradient and reflecting rapid vertical mixing upon alkalinity addition (Supplementary Fig. 4b).

After 48 h (on D6), all mesocosms were fully mixed, simulating dilution of high-alkalinity waters with untreated seawater (Fig. 1b). After mixing, ΔTA levels were expected to match in intensity across both delayed and immediate dilution treatments, ensuring an equivalent ΔTA gradient for direct comparison. The final realised ΔTA after mixing deviated by no more than 70 $\mu\text{mol kg}^{-1}$ between dilution treatments of the same ΔTA level

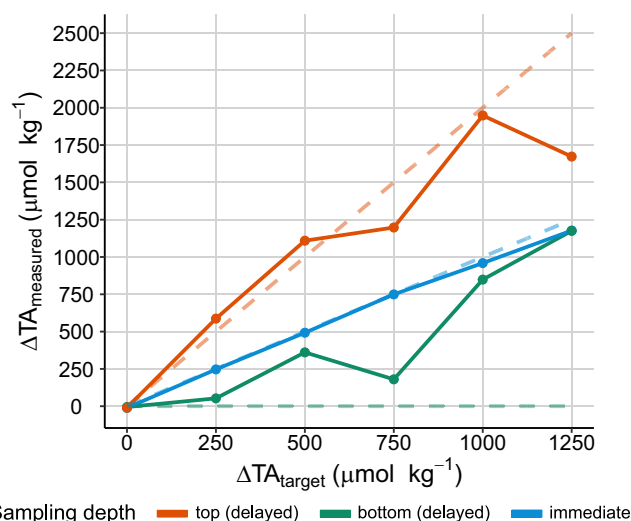


Fig. 2 | Targeted versus measured ΔTA during days 5 and 6. Targeted ΔTA ($\Delta TA_{\text{target}}$; $\mu\text{mol kg}^{-1}$) versus average (D5 and D6) measured ΔTA ($\Delta TA_{\text{measured}}$; $\mu\text{mol kg}^{-1}$) in the top (orange) and bottom layer (green) of the delayed and immediate dilution (blue) treatments. Dashed and full lines indicate the target and measured ΔTA , respectively, for each dilution treatment.

(Fig. 3a). Detailed values of target ΔTA , measured ΔTA on D7 and average measured ΔTA (D7–D39), as well as the corresponding fugacity of CO_2 ($f\text{CO}_2$) levels, are provided in Supplementary Table 1. ΔTA remained stable across all mesocosms throughout the 39-days-long experiment, with the exception of the highest ΔTA treatment (target ΔTA 1250 $\mu\text{mol kg}^{-1}$), where TA loss was observed. After mixing the water column, initial ΔTA levels in the highest ΔTA treatment reached $\sim 1150 \mu\text{mol kg}^{-1}$ in both dilution treatments on D7. However, abiotic precipitation in the highest ΔTA treatment was detected from D17 onwards in the delayed and from D23 in the immediate dilution treatment and continued until the end of the experiment. By D39, measured ΔTA had dropped to $\sim 115 \mu\text{mol kg}^{-1}$ (delayed) and $\sim 500 \mu\text{mol kg}^{-1}$ (immediate), corresponding to a TA loss of ~ 90 and $\sim 55\%$ relative to the initial alkalinities reached on D7 after water column mixing. In these two treatments, the TA loss is indicative of secondary abiotic CaCO_3 precipitation as supported by a concurrent decrease in DIC (Supplementary Fig. 7a).

A pronounced decline in $f\text{CO}_2$ occurred on D4 across all mesocosms after alkalinity enhancement, as expected from alkalinity addition (Fig. 3b). Despite substantial TA loss and a decrease in pH in the highest ΔTA treatment due to abiotic precipitation, no corresponding increase in $f\text{CO}_2$ was observed. This likely reflects the buffering capacity provided by the initially elevated alkalinity, which reduced the sensitivity of $f\text{CO}_2$ to carbonate chemistry changes, together with the simultaneous removal of DIC during CaCO_3 precipitation. As a result, the effects of TA loss and decreasing pH on $f\text{CO}_2$ were offset by the concurrent decline in DIC (Supplementary Fig. 7a). Notably, in the controls (no added alkalinity), a decline in $f\text{CO}_2$ was observed, which can be attributed to CO_2 consumption during photosynthetic carbon fixation. In contrast, this biologically driven signal in $f\text{CO}_2$ was likely buffered in the alkalisied mesocosms, with potentially additional moderation from partial atmospheric CO_2 in-gassing, thereby suppressing changes in $f\text{CO}_2$ despite ongoing photosynthesis as seen in Chl *a*. Nevertheless, across the full experimental period, mesocosms with target ΔTA additions of $\geq 750 \mu\text{mol kg}^{-1}$ persistently maintained $f\text{CO}_2$ values below $58 \mu\text{atm}$ (Fig. 3b).

For defining biological thresholds based on ΔTA , we refer to measured ΔTA and $f\text{CO}_2$ levels from D7 ($\Delta\text{TA}_{\text{measured}}$; Supplementary Table 1), as this represents the point at which carbonate chemistry conditions were fully established after manipulation and mixing. Although ΔTA eventually declined in the highest ΔTA treatment due to secondary precipitation, $f\text{CO}_2$ remained comparatively stable. Thus, the initial ΔTA manipulation effectively defined the $f\text{CO}_2$ conditions to which the phytoplankton were

exposed to over the course of the experiment. Because we assume $f\text{CO}_2$ to be the main driver of phytoplankton responses, reporting the D7 measured ΔTA values provides the most consistent representation of the carbonate chemistry treatment conditions relevant for biological interpretations. Measured values from D5 and D6 are reported where relevant to describe the two-layer phase, while the target ΔTA levels are used for orientation in figures and clearly marked when used in the text ($\Delta\text{TA}_{\text{target}}$).

Dissolved inorganic nutrient utilization

Highest concentrations of NO_x ($\text{NO}_3 + \text{NO}_2$), PO_4 and $\text{Si}(\text{OH})_4$ were measured at the very first day of the experiment ($\text{NO}_x = 18.8 \pm 0.4 \mu\text{mol L}^{-1}$, $\text{PO}_4 = 0.5 \pm 0.0 \mu\text{mol L}^{-1}$, $\text{Si}(\text{OH})_4 = 14.5 \pm 0.1 \mu\text{mol L}^{-1}$), followed by a progressive decrease (Fig. 4). Notably, across both dilution treatments, the high ΔTA mesocosms ($\Delta\text{TA}_{\text{target}} \geq 1000 \mu\text{mol kg}^{-1}$; $\Delta\text{TA}_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$) exhibited slower nutrient declines compared to the low to moderate ΔTA treatments ($\Delta\text{TA}_{\text{target}} \leq 750 \mu\text{mol kg}^{-1}$; $\Delta\text{TA}_{\text{measured}} \leq 700 \mu\text{mol kg}^{-1}$), pointing towards an alkalinity mediated delay in phytoplankton nutrient uptake in these treatments.

Phosphate was the primary limiting nutrient, showing a rapid decline in the low to moderate ΔTA treatments ($\Delta\text{TA}_{\text{measured}} \leq 700 \mu\text{mol kg}^{-1}$) relative to the higher ΔTA treatments ($\Delta\text{TA}_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$), and reaching levels below the detection limit by D11 (Fig. 4a). The higher ΔTA mesocosms ($\Delta\text{TA}_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$) in both dilution treatments retained measurable phosphate for longer, reaching concentrations below detection between D13 and D29.

Fig. 3 | Temporal development of carbonate chemistry throughout the experiment. a ΔTA ($\mu\text{mol kg}^{-1}$) and **b** $f\text{CO}_2$ (μatm) over the course of the experiment (39 days). Purple and red vertical lines indicate alkalinity addition on D4 and mixing of the water column on D6, respectively. Blue shades, open symbols and solid lines represent the immediate dilution treatment, green shades, filled symbols and dashed lines the delayed dilution treatment. During the two-layer phase (D5 and D6), the top layer of the delayed dilution treatment is shown in orange shades and the bottom layer in green shades. Colour intensity corresponds to increasing alkalinity.

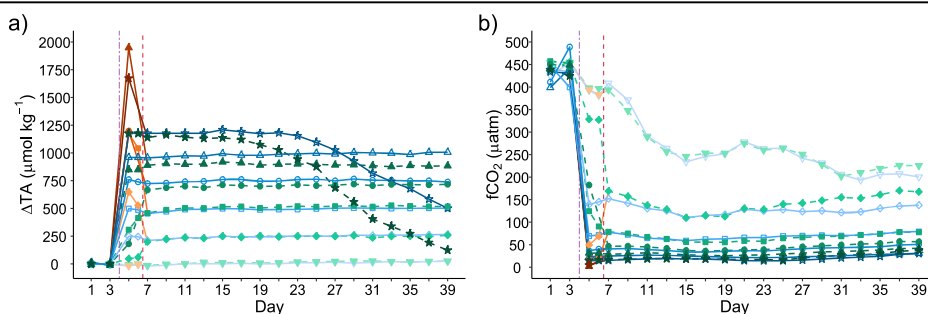
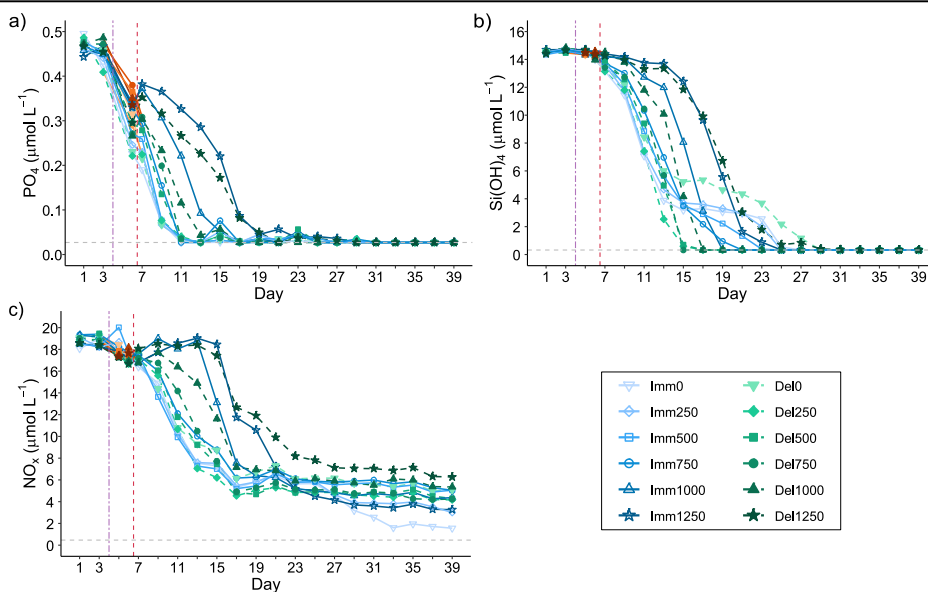


Fig. 4 | Temporal development of dissolved inorganic nutrient concentrations. a PO_4 , **b** $\text{Si}(\text{OH})_4$ and **c** NO_x concentrations ($\mu\text{mol L}^{-1}$) over the course of the experiment (39 days). Vertical purple and red lines indicate days of alkalinity addition (D4) and water column mixing (D6), respectively. The delayed dilution treatment is represented in green with dashed lines and filled symbols, while the immediate dilution treatment is shown in blue with solid lines and open symbols. On D5 and D6, measurements from the top layer of the delayed dilution treatments are shown in orange shades and from the bottom layer in green shades. Values reported in the figure legend refer to post-mixing $\Delta\text{TA}_{\text{target}}$ levels ($\mu\text{mol kg}^{-1}$; Imm immediate, Del delayed dilution treatment). Horizontal grey dashed lines indicate the detection limit of each nutrient. Values at or below the detection limit (LOD) are shown at the LOD value (LOD: $\text{PO}_4 = 0.027 \mu\text{mol L}^{-1}$, $\text{Si}(\text{OH})_4 = 0.323 \mu\text{mol L}^{-1}$, $\text{NO}_x = 0.467 \mu\text{mol L}^{-1}$).



Silicic acid dynamics revealed potential complex interactions between alkalinity levels and $\text{Si}(\text{OH})_4$ uptake by phytoplankton. $\text{Si}(\text{OH})_4$ concentrations displayed differences in depletion timing across alkalinity treatments. In both dilution treatments, earlier and relatively rapid $\text{Si}(\text{OH})_4$ depletion occurred in those mesocosms of low to moderate ΔTA . In contrast, $\text{Si}(\text{OH})_4$ depletion stagnated under high ΔTA conditions (until D15 at $\Delta\text{TA}_{\text{target}} 1250 \mu\text{mol kg}^{-1}$, Fig. 4b). In the mesocosms with $\Delta\text{TA}_{\text{measured}} \leq 700 \mu\text{mol kg}^{-1}$ in the immediate dilution treatment as well as in the control of the delayed dilution treatment, a two-step decline was observed, starting with a steep drop in $\text{Si}(\text{OH})_4$ concentrations from D7 to D13, followed by a gradual decline from D17 to D29 when $\text{Si}(\text{OH})_4$ became depleted in all mesocosms.

The dynamics of NO_x showed a distinct pattern. Similar to $\text{Si}(\text{OH})_4$ dynamics, NO_x decline during the first 15 days of the experiment was delayed in treatments with $\Delta\text{TA}_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$ relative to the lower ΔTA treatments. Notably, NO_x concentrations in all mesocosms remained above $1 \mu\text{mol L}^{-1}$ throughout the experiment and were never fully depleted (Fig. 4c).

Temporal development of Chl *a*

Prior to alkalinity addition, all mesocosms exhibited increasing Chl *a* concentrations, indicative of good timing of the experiment to capture the spring bloom development. Note that ΔTA values reported here refer to the $\Delta\text{TA}_{\text{measured}}$ levels during the two-layer phase (D5 and D6; Figs. 2 and 5). Following alkalinity addition and two-layer establishment (D5 and D6), the

increase in Chl *a* concentrations in the top layer of the delayed dilution treatment was increasingly suppressed with alkalinity, indicating a clear dose-dependent response (Fig. 5). Additionally, in the delayed dilution treatment, Chl *a* concentrations at top layer $\Delta\text{TA}_{\text{measured}} \geq 1000 \mu\text{mol kg}^{-1}$ were lower in the alkalised top layer relative to the bottom layer of the same mesocosm (Fig. 5c–f). These responses in the delayed dilution treatments are further supported by the average change in Chl *a* during the two-layer phase (D5–D6) relative to the average baseline concentrations on D1–D3 ($\Delta\text{Chl } a$, Supplementary Fig. 8), indicating lower average Chl *a* concentrations in the top layers at $\Delta\text{TA}_{\text{measured}} \geq 1000 \mu\text{mol kg}^{-1}$ compared to lower ΔTA treatments.

Among delayed dilution treatments, only the top layer treated with $\Delta\text{TA}_{\text{target}} 500 \mu\text{mol kg}^{-1}$ ($\Delta\text{TA}_{\text{measured}} 587 \mu\text{mol kg}^{-1}$, $f\text{CO}_2 \sim 60 \mu\text{atm}$, Fig. 5b) showed Chl *a* development similar to the control. This indicates the higher end of the $f\text{CO}_2$ threshold below which Chl *a* in the top layer was suppressed. Notably, the bottom layers of high alkalinity treatments (Fig. 5d–f), which were designed to remain unaffected, also showed lower Chl *a* concentrations relative to the controls, indicating unintended sampling of the transition zone between the alkalised top layers and bottom layers where alkalinity was not added.

Chl *a* concentrations also declined in the immediate dilution treatment at $\Delta\text{TA}_{\text{measured}} \geq 1000 \mu\text{mol kg}^{-1}$, although the decrease was slightly less pronounced (Fig. 5e, f). Corresponding $f\text{CO}_2$ levels were $23.7 \pm 0.7 \mu\text{atm}$ (immediate, Fig. 5e) and $18.1 \pm 4.5 \mu\text{atm}$ (delayed top layer, Fig. 5c) for $\Delta\text{TA}_{\text{measured}}$ of $\sim 1000 \mu\text{mol kg}^{-1}$ (Fig. 2 and Supplementary Fig. 8).

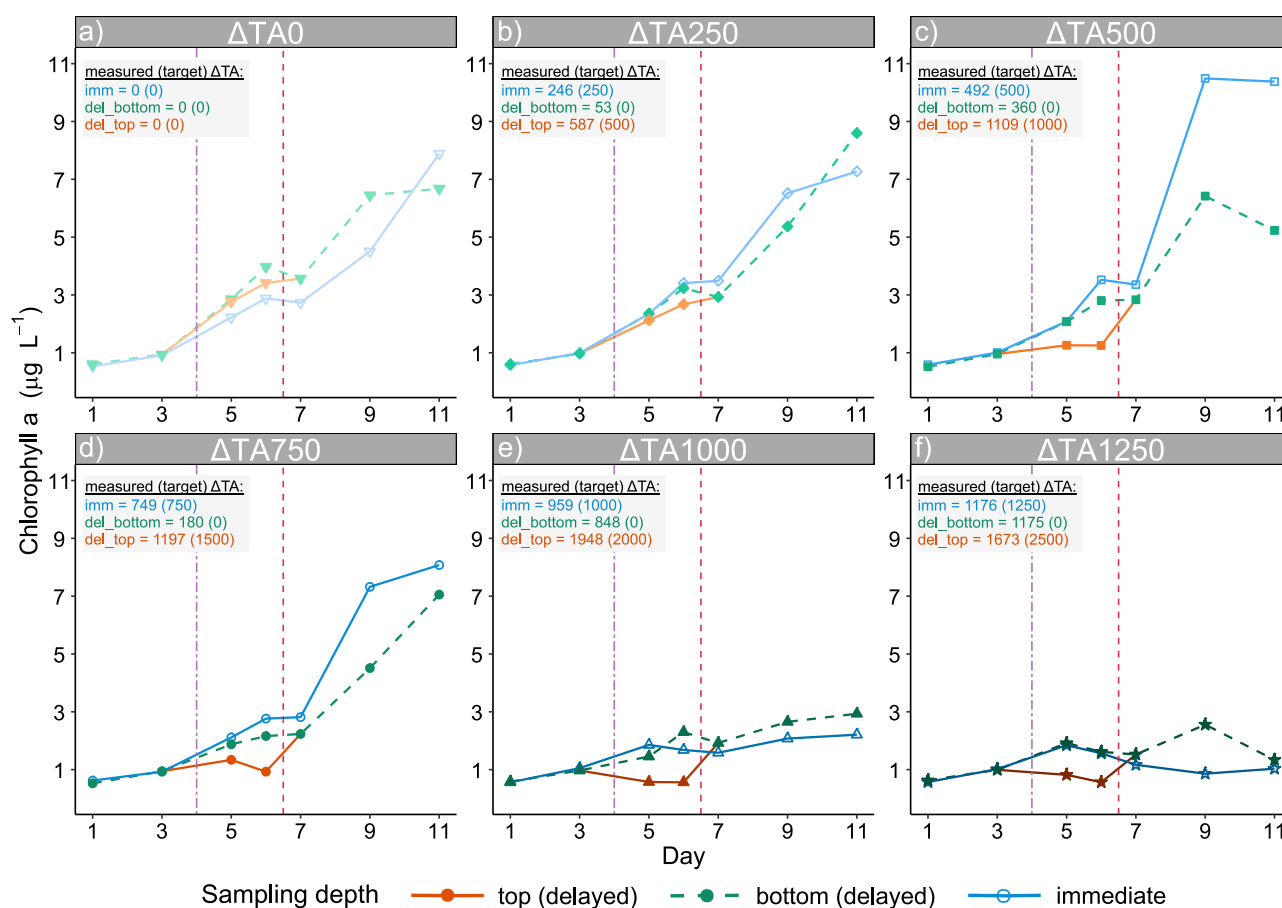


Fig. 5 | Chlorophyll *a* dynamics from day 1 through the two-layer phase and early post-mixing phase. a–f Close up of the chlorophyll *a* concentration ($\mu\text{g L}^{-1}$) during the two-layer phase of OAE manipulation on D5 and D6, divided into the $\Delta\text{TA}_{\text{target}}$ levels after mixing (D7) from 0 to $1250 \mu\text{mol kg}^{-1}$. The immediately diluted treatment is shown in blue (solid lines, open symbols), while the delayed dilution treatment is separated into the top layer (orange, solid lines, filled symbols) and the bottom layer (green, dashed lines, filled symbols). Colour intensity increases with

ΔTA . Purple and red vertical lines indicate alkalinity addition on D4 and mixing of the water column on D6, respectively. Panel titles indicate targeted ΔTA after full water column mixing (from D7 onwards). Values shown in grey boxes within each panel represent mean measured ΔTA ($\mu\text{mol kg}^{-1}$) during the two-layer phase on D5 and D6, with targeted ΔTA levels ($\mu\text{mol kg}^{-1}$) given in parentheses for each treatment and layer.

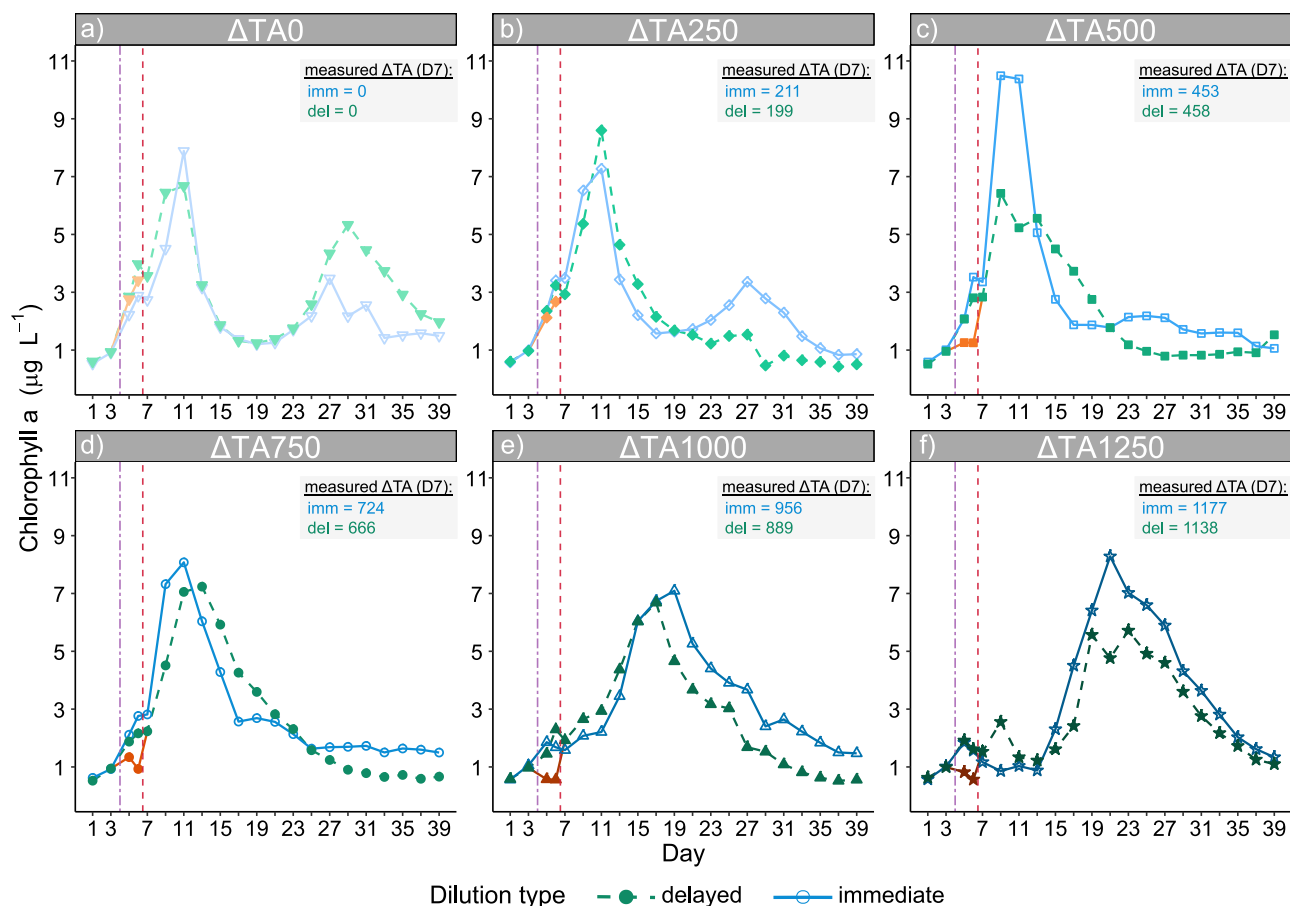


Fig. 6 | Chlorophyll *a* dynamics over the full experimental period. a–f Chlorophyll *a* concentrations (µg L⁻¹) over the course of the experiment (39 days) divided into the ΔTA_{target} levels from 0 to 1250 µmol kg⁻¹. This enables the comparison of the two dilution scenarios. Values shown in grey boxes within each panel represent ΔTA_{measured} (µmol kg⁻¹) on D7. Blue and green shades represent immediate (open symbols) and delayed (filled symbols) dilution treatments, respectively. On D5 and

D6, measurements from the top layer of the delayed dilution treatment are shown in orange shades and from the bottom layer in green shades. Purple and red vertical lines indicate alkalinity addition on D4 and mixing of the water column on D6, respectively. Colour intensity from light to dark represents increasing intensity of alkalinity added.

After the mixing on D6, Chl *a* dynamics revealed distinct phytoplankton bloom development patterns (Fig. 6) consistent with the decline in dissolved inorganic nutrients (Fig. 4). The observed responses to OAE were nonlinear, and obvious negative effects on bloom development only became apparent once alkalinity has been added beyond a certain level. In low to intermediate ΔTA levels (ΔTA_{measured} ≤ ~700 µmol kg⁻¹, fCO₂ ≥ 41 µatm; Supplementary Table 1 and Fig. 6a–d), the exponential phytoplankton growth continued after the alkalinity addition and mixing, reaching peak average Chl *a* concentrations of 7.8 (± 1.3) µg L⁻¹ between D9 and D13. In contrast, the two highest ΔTA treatments (ΔTA_{measured} ≥ ~900 µmol kg⁻¹, fCO₂ ≤ 29 µatm, Fig. 6e, f) of both dilution treatments displayed different bloom patterns, characterized by a lag phase before Chl *a* increase. To quantify and compare bloom development across ΔTA treatments and dilution types, we use the day of maximum exponential growth as a common biological reference point. For this, we calculated the daily rate of change of nine different biogeochemical parameters (NO₃, NO₂, Si(OH)₄, PO₄, Chl *a*, Biogenic Silica (BSi), particulate organic carbon (POC), nitrogen (PON) and phosphorus (POP)) and identified, for each parameter, the day with the largest rate of change (i.e., maximum rate of growth for Chl *a* and POM components and maximum rate of decline for inorganic nutrients). For each mesocosm, kernel density estimates (KDE) were constructed from these nine parameter-specific days of maximum rate of change. The mode of each distribution was taken as the representative day of maximum exponential growth, representing a well-defined and objectively determined point in bloom development at which the phytoplankton community is in a

comparable physiological state across all mesocosms (see the “Methods” section, Supplementary Fig. 9 and Supplementary Table 2). Quantitative analysis, based on this definition of a key bloom development stage post-mixing, revealed a strong relationship between fCO₂ and phytoplankton bloom development (Fig. 7). Alkalinity levels of ΔTA_{measured} ≤ ~700 µmol kg⁻¹ (fCO₂ ≥ 41 µatm), showed no significant differences in timing of maximum exponential growth (~D8, linear regression with R² = 0.029 and p = 0.685). In contrast, ΔTA_{measured} levels ≥ ~900 µmol kg⁻¹ (fCO₂ ≤ 29 µatm) exhibited a pronounced delay with maximum exponential growth occurring ~6–9 days later relative to the ΔTA_{measured} additions ≤ ~700 µmol kg⁻¹ (Fig. 7 and Supplementary Table 2).

We mapped out the parameter-specific estimates for the day of maximum exponential growth, reflected by the 10th–90th percentile error bars (Fig. 7). This spread is expected because the KDE distribution integrated nine biologically distinct estimates of maximum rates of change, which naturally do not occur on exactly the same day. Rather than representing uncertainty around the mode, the spread reflects natural parameter-specific differences in bloom-related response timing. The KDE-derived mode, therefore, remains the most representative summary estimate of the timing of maximum exponential growth for each mesocosm. A Mann–Whitney *U* test indicated a significant difference with no overlap (*U* = 0, *n*₁ = 4, *n*₂ = 8, *p* = 0.008; *n* = mesocosms per group) in the modal estimates of the day of the maximum exponential growth between low to moderate (ΔTA_{measured} ≤ ~700 µmol kg⁻¹, fCO₂ ≥ 41 µatm) and high (ΔTA_{measured} ≥ ~900 µmol kg⁻¹, fCO₂ ≤ 29 µatm) alkalinity treatments. This threshold

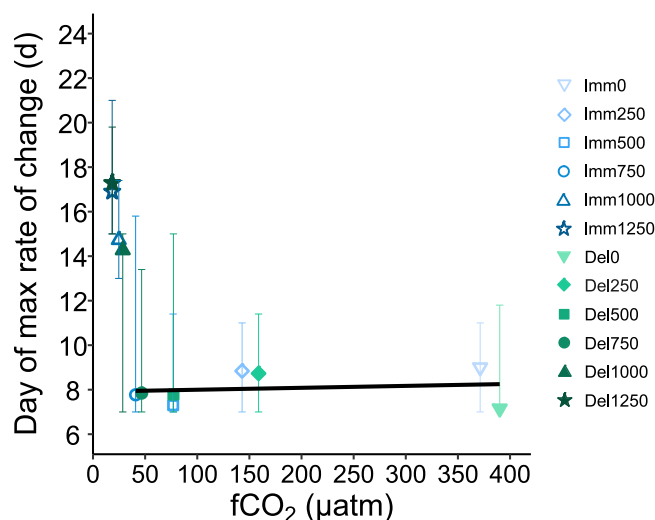


Fig. 7 | Bloom development timing along the $f\text{CO}_2$ gradient. Relationship between seawater $f\text{CO}_2$ (μatm) and KDE-derived bloom development timing (d). Symbols represent the modal estimate (i.e., the mode of the KDE distribution, used as the measure of centre) for the day of maximum rate of change across nine biological variables for each mesocosm ($n = 9$ biological measurements). The vertical bars denote the 10th and 90th percentile range of the nine parameter-specific estimates for each mesocosm. This range reflects the temporal spread of biological responses, integrating diverse observations with maximum rates of change that vary because of stoichiometric differences. Thus, the range captures differences between responding variables rather than reflecting uncertainty arising from stochastic variability around the mode. Dilution types are separated by colour shades (green, filled symbols = delayed dilution; blue, open symbols = immediate dilution). Increasing colour intensity corresponds to increasing ΔTA . The solid black line shows the linear regression fitted to the modal estimates for treatments with $\Delta\text{TA}_{\text{measured}}$ levels $\leq \sim 700 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \geq 41 \mu\text{atm}$; $R^2 = 0.029$, $p = 0.685$). The Mann–Whitney U test comparing modal estimates between lower ($\Delta\text{TA}_{\text{measured}} \leq \sim 700 \mu\text{mol kg}^{-1}$, $f\text{CO}_2 \geq 41 \mu\text{atm}$) and higher ($\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$, $f\text{CO}_2 \leq 29 \mu\text{atm}$) ΔTA treatment groups revealed a significant difference between groups ($U = 0$, $n_1 = 4$, $n_2 = 8$, $p = 0.008$; n = mesocosms per group). Values reported in the figure legend refer to post-mixing $\Delta\text{TA}_{\text{target}}$ levels ($\mu\text{mol kg}^{-1}$; Imm immediate, Del delayed dilution treatment).

response to ΔTA and $f\text{CO}_2$ was independent of the type of dilution treatments. A similar relationship was observed between proton concentrations (H^+) and phytoplankton growth response (Supplementary Fig. 10).

In addition to a delay in phytoplankton bloom development, alkalinity also influenced the duration of the bloom and, consequently, the dynamics of bloom decay. Under low to moderate alkalinity ($\Delta\text{TA}_{\text{measured}} \leq \sim 700 \mu\text{mol kg}^{-1}$, Fig. 6a–d), total Chl a concentrations showed steep declines following the bloom peak. In contrast, higher alkalinity treatments ($\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$, Fig. 6e, f) showed more gradual declines. This trend is explicitly captured in the rates of change of Chl a and Chl a -specific bloom length, defined as the time between the day of maximum rate of change of Chl a and the day the rate of change returns to zero (Supplementary Fig. 11 and Supplementary Table 3). Chl a -specific bloom length, which closely tracks phytoplankton dynamics, increased from ~ 11 days in the control mesocosms to ~ 23 days in the mesocosm with the highest ΔTA additions ($\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$, Supplementary Table 3). In the highest ΔTA treatment ($\Delta\text{TA}_{\text{target}} 1250 \mu\text{mol kg}^{-1}$), the Chl a bloom duration was likely even longer; however, due to the delayed bloom development in these mesocosms, the experiment ended before the rate of change of Chl a returned to zero.

Furthermore, the temporal development of Chl a concentrations revealed a secondary, less pronounced phytoplankton bloom in both control mesocosms ($\Delta\text{TA}_{\text{target}} 0 \mu\text{mol kg}^{-1}$) and in the immediate dilution treatment at $\Delta\text{TA}_{\text{measured}} \sim 200 \mu\text{mol kg}^{-1}$ (Fig. 6a, b). These secondary blooms were characterised by both Chl a increases and distinct $\text{Si}(\text{OH})_4$ utilization

rates (Fig. 4 and Supplementary Fig. 12). Although $\text{Si}(\text{OH})_4$ dynamics suggest potential secondary blooms in the immediate dilution treatment with $\Delta\text{TA}_{\text{measured}} \sim 450$ and $\sim 700 \mu\text{mol kg}^{-1}$, these patterns were not clearly reflected in Chl a concentrations (Fig. 6c, d) and we therefore do not classify these as definitive secondary blooms. No secondary Chl a blooms were observed in any delayed dilution treatment at $\Delta\text{TA}_{\text{measured}} \geq \sim 200 \mu\text{mol kg}^{-1}$ or in the highest immediate dilution treatments ($\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$; Fig. 6e, f), coinciding with $\text{Si}(\text{OH})_4$ concentrations below the detection limit by the end of the primary blooms.

Suspended particulate organic matter dynamics and stoichiometry

Temporal dynamics of all suspended POM components (POC, PON and POP) and BSi closely followed Chl a development over the course of the experiment (Supplementary Figs. 13–17). BSi concentrations strongly followed Chl a dynamics, indicating a significant contribution of diatoms to total Chl a concentrations (Supplementary Figs. 13 and 14). This is further supported by the temporal dynamics of Fucoxanthin, the primary accessory pigment for diatoms, which followed the development of Chl a (Supplementary Fig. 18). Furthermore, a strong positive correlation between Chl a and Fucoxanthin was identified (Supplementary Fig. 19), whereby the variance in fucoxanthin concentrations explained 85% of Chl a variation, suggesting that fucoxanthin-containing phytoplankton (primarily diatoms) were significant contributors to the total phytoplankton biomass across all treatments and timepoints.

In contrast, the temporal dynamics of POC, POP and PON differed from those of Chl a and BSi (Supplementary Figs. 15–17). While Chl a and BSi concentrations dropped sharply during the bloom collapse, the decline in POC, POP and PON was more gradual and less pronounced. This likely reflects the broader biological origin of the specific POM pools. While BSi and Chl a are tightly coupled to phytoplankton dynamics, POC, PON and POP reflect broader biological processes beyond phytoplankton, which include contributions from zooplankton and detritus. As a result, POC, PON and POP also capture the potential retention of organic matter in the water column relative to sinking during the bloom decline and nutrient limited phase of the experiment.

In addition to shifts in bloom timing, changes in POM components provide insights into how OAE scenarios may affect biomass present during distinctive bloom periods. As an indicator of magnitude and for comparison, temporal averages of the individual POM components, including Chl a and BSi, were determined. For averaging, the time period between the maximum exponential growth and the end of the bloom, defined as the time when the rate of change across parameters returned to zero, was chosen independently for each alkalinity treatment. This averaging window does not capture the full bloom period, but focuses on two clearly defined key bloom development stages within each alkalinity treatment. It therefore represents distinct and comparable states of the plankton ecosystem and provides a biologically comparable basis for cross-treatment comparisons, despite OAE-induced differences in bloom timing.

To identify treatment effects, we employed an empirical relationship adapted from Gafar et al.⁴⁴ to describe the dependence of the observed average concentrations on seawater $f\text{CO}_2$. The nonlinear model was fitted using bootstrap-resampled datasets. Confidence intervals derived from the resulting ensemble of model fits constrain the range of plausible solutions, with their spread reflecting the variability in the observations. Model outputs were further evaluated against a no-effect model, representing the mean of the original data. A detectable OAE response was inferred when the 10th–90th percentile range of ensemble residuals did not include zero (Supplementary Fig. 20). This framework enables the detection of significant dependencies on $f\text{CO}_2$.

Overall, our analysis shows no pronounced differences in average biomass in response to most OAE treatments (Fig. 8 and Supplementary Fig. 20). Compared with Chl a and other POM components, POP appeared most sensitive to low $f\text{CO}_2$ levels, in both dilution treatments (Fig. 8i,j). At

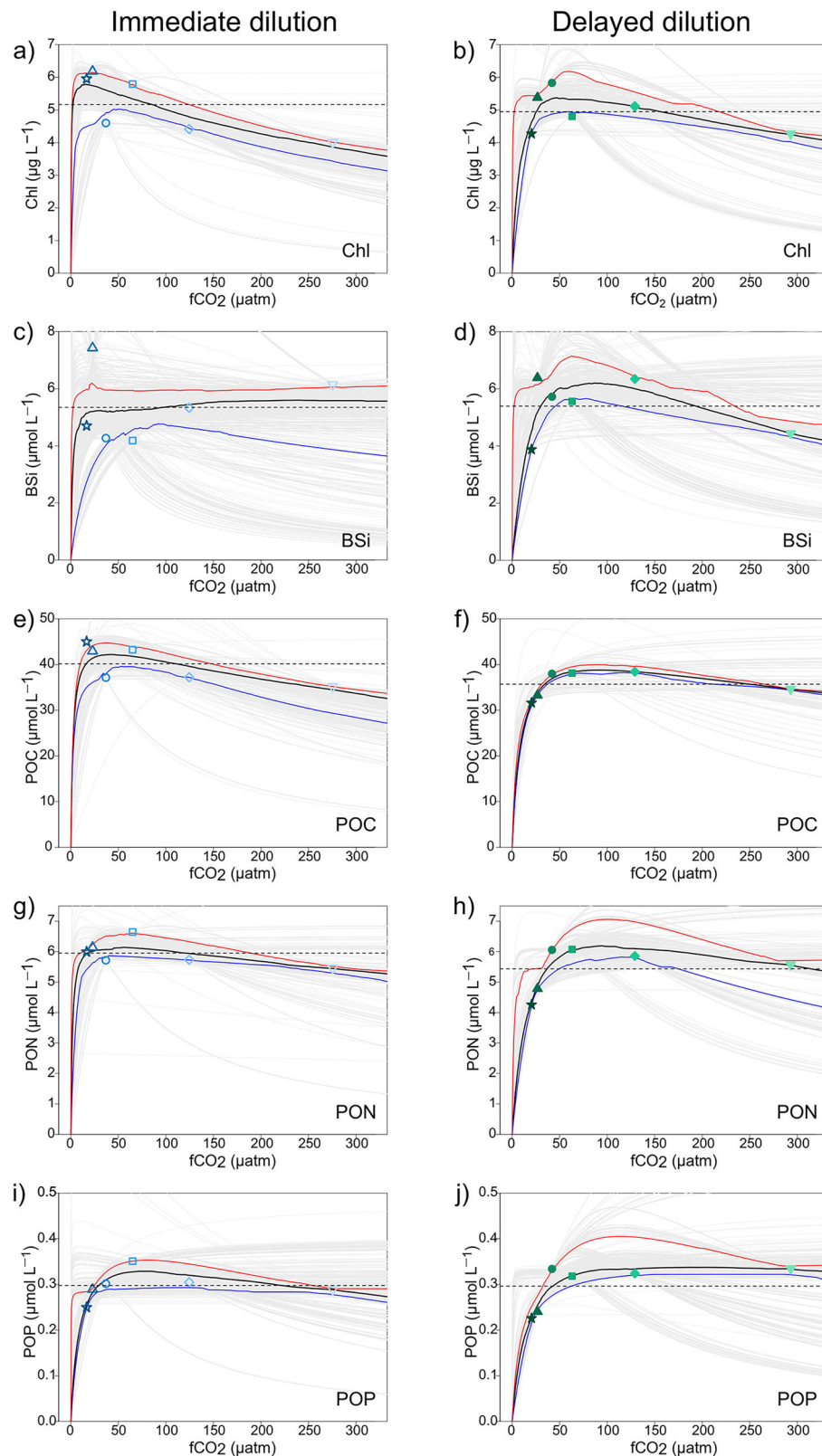


Fig. 8 | Biogeochemical responses across $f\text{CO}_2$ levels. Relationship between seawater $f\text{CO}_2$ (μatm) and average concentrations of Chl *a* (a, b; $\mu\text{g L}^{-1}$), BSi (c, d; $\mu\text{mol L}^{-1}$), POC (e, f; $\mu\text{mol L}^{-1}$), PON (g, h; $\mu\text{mol L}^{-1}$) and POP (i, j; $\mu\text{mol L}^{-1}$) during the key bloom development phase (from the day of maximum exponential growth to the end of the bloom). The immediate dilution treatment is represented on the left-hand side (a, c, e, g, i) while the delayed dilution treatment is shown on the right-hand side (b, d, f, h, j). The nonlinear model was adapted from Gafar et al.⁴⁴ with grey lines

representing the ensemble of model solutions (2000). The red and blue lines show the upper and lower confidence limits (10th and 90th percentile), while the black line represents the median (50th percentile) of the ensemble of the model solutions. The horizontal black line indicates the no-effect model, which is equal to the mean of the original observations. Dilution types are separated by colour shades (green = delayed dilution, blue = immediate dilution). Increasing colour intensity corresponds to increasing TA.

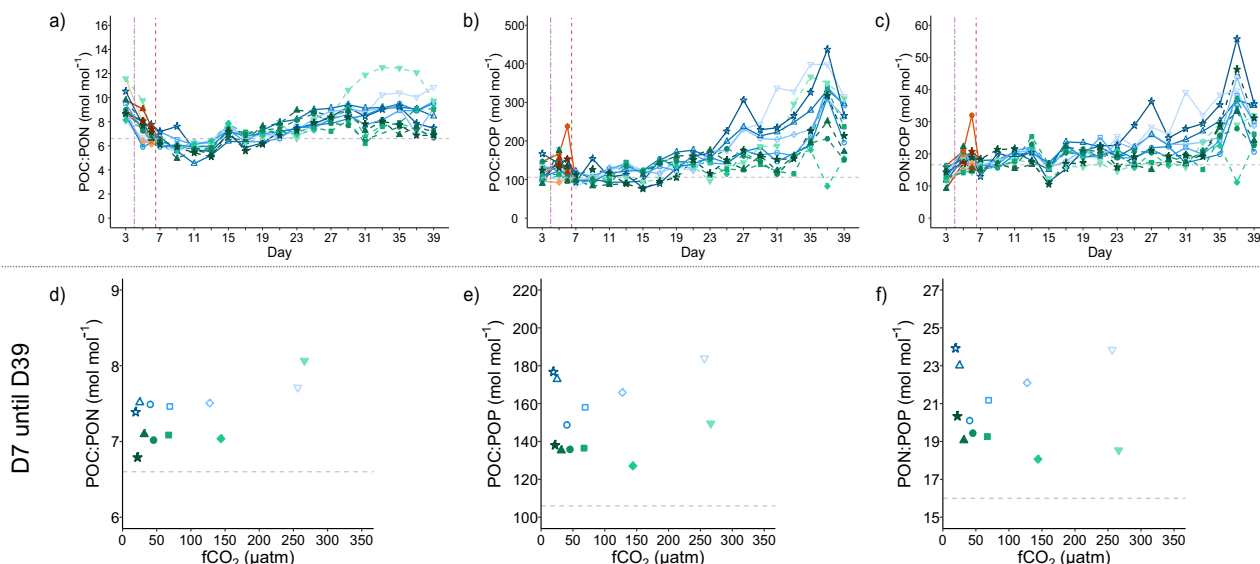


Fig. 9 | Elemental stoichiometry across time and $f\text{CO}_2$ levels. a–c Temporal development of elemental ratios POC:PON (a), POC:POP (b) and PON:POP (c; mol mol⁻¹) over the course of the experiment and d–f relationship of geometric means of elemental ratios with $f\text{CO}_2$ (POC:PON (d), POC:POP (e) and PON:POP (f; mol mol⁻¹)) from D7 to the end of the experiment (D39). Grey dashed lines show Redfield ratios for POC:PON = 6.6, POC:POP = 106 and PON:POP = 16,

respectively. Dilution types are separated by colour shades (green, filled symbols = delayed dilution, blue, open symbols = immediate dilution). On D5 and D6, ratios from the top layer of the delayed dilution treatment are shown in orange shades and from the bottom layer in green shades. Purple and red vertical lines indicate alkalinity addition on D4 and mixing of the water column on D6, respectively. Increasing colour intensity corresponds to increasing ΔTA .

the lowest $f\text{CO}_2$ level, average POP concentrations were ~23% lower in the delayed dilution (0.23 μmol kg⁻¹ vs. 0.3 μmol kg⁻¹) and ~15% lower in the immediate dilution treatment (0.25 μmol kg⁻¹ vs. 0.3 μmol kg⁻¹) relative to the mean of the original data (no-effect model). For POC and PON, $f\text{CO}_2$ effects were detected only in the delayed dilution treatment with up to ~11 and ~21% lower values, respectively, relative to the no-effect model (31.72 vs. 35.69 μmol kg⁻¹ for POC; 4.27 vs. 5.43 μmol kg⁻¹ for PON, Fig. 8f, h). Notably, although derived independently, the estimated $f\text{CO}_2$ threshold levels were comparable in all four cases, ranging from 25 to 32 μatm, corresponding to $\Delta\text{TA}_{\text{measured}}$ of ≥ 900 μmol kg⁻¹ in this study (Supplementary Fig. 20f, h–j and Supplementary Table 1). In both dilution treatments, average BSi and Chl *a* concentrations during the assessed bloom period showed high variability across the $f\text{CO}_2$ gradient, thus no significant $f\text{CO}_2$ response was detected.

Elemental stoichiometric ratios were assessed over the full post-mixing period (D7 until D39) to capture the complete succession of organic matter quality following OAE manipulation. This extended window is appropriate because elemental ratios reflect the cumulative contributions of multiple biological sources, including phytoplankton, zooplankton, heterotrophic bacteria and detritus, whose relative contributions shift substantially over the bloom cycle. Furthermore, OAE-induced effects on nutrient cycling and community succession may manifest as gradual stoichiometric shifts over longer timescales that would not be captured within the bloom phase window alone.

Temporal trends of POC:PON, POC:POP and PON:POP did not reflect the delay in phytoplankton bloom development observed at high ΔTA levels ($\Delta\text{TA}_{\text{measured}} \geq 900$ μmol kg⁻¹, Fig. 9a–c). Across treatments, stoichiometric ratios generally remained close to Redfield values (6.6 for POC:PON, 106 for POC:POP, 16 for PON:POP), with lower values during the first half of the experiment and elevated values during the second half, coinciding with the period of phosphate limitation (Fig. 4a). Notably, the secondary bloom observed in the control mesocosms coincided with elevated POC:PON ratios relative to the alkalinised treatments towards the end of the experiment as shown in the temporal development (Fig. 9a).

Furthermore, during our experiment, stoichiometric ratios showed high variability across treatments and dilution types (Fig. 9d–f). Thus, no

consistent dose-dependent relationship between mean elemental ratios over the post-mixing experimental period (D7 until D39) and alkalinity (and $f\text{CO}_2$) was detected across both dilution treatments (Fig. 9d–f). Notably, the elevated POC:PON ratios associated with the secondary bloom during the last third of the experiment likely raised the mean ratio (D7 until D39) in the delayed dilution control. Accordingly, this control mesocosm showed the highest mean POC:PON value across the full experimental period, exceeding all other treatments (average POC:PON > 8, Fig. 9d). In contrast, in the immediate dilution control, averaging over the post-mixing experimental period smoothed out this temporal signal, resulting in mean POC:PON values comparable to the other alkalinity treatments (average POC:PON ~ 7–8).

Discussion

We conducted an in-situ mesocosm experiment in the temperate-eutrophic waters of the German North Sea to assess how OAE influences planktonic ecosystem dynamics over short- and medium-term timescales. The experimental design contrasts two OAE exposure modes simulating hydrated lime deployment in well-mixed alkalinity additions (immediate dilution) versus concentrated surface-layer additions with delayed mixing (delayed dilution). The delayed dilution treatment simulates deployment conditions where alkaline inputs remain stratified for some time before vertical or lateral dispersion. In this case, ambient phytoplankton communities in the top 0.8 m layer of the mesocosms were exposed to a gradient of high alkalinity for 48 h before fully mixing the water column, diluting alkalinity by half to match the alkalinity level in the immediate dilution treatment. Accordingly, we tested whether short-term perturbations are ecologically disruptive and whether communities can recover once alkalinised waters are diluted.

Short-term effects of alkalinity addition on Chl *a*

The short-term OAE response in this study, defined as the immediate period following alkalinity addition and two-layer establishment (D5 and D6), was reflected in the temporal development of Chl *a* concentrations, which were clearly lower at high alkalinity levels, indicating changes in phytoplankton growth and photosynthetic activity (Fig. 5 and Supplementary Fig. 8). In the

top layer of the delayed dilution treatment, where alkalinity was added only to the upper 0.8 m, resulting in double the post-mixing target ΔTA , a clear reduction in Chl *a* was evident at $\Delta TA_{\text{measured}} \geq \sim 1000 \mu\text{mol kg}^{-1}$ (Fig. 5c–f; values refer to the average $\Delta TA_{\text{measured}}$ levels in the top layer at D5 and D6, Fig. 2). A similar threshold response for lower Chl *a* concentrations was observed for the immediate dilution treatment at equivalent $\Delta TA_{\text{measured}} \geq \sim 1000 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \leq 23 \mu\text{atm}$). The low concentrations in Chl *a*, observed within a 48-h window after alkalinity addition, support our initial hypothesis that high alkalinity exposure impairs phytoplankton growth and demonstrate sensitivity and rapid response of the phytoplankton community to acute perturbations in the carbonate system.

Alkalinity additions strongly reduced $f\text{CO}_2$ and raised pH, creating conditions known to limit carbon availability for phytoplankton⁴⁵. At high alkalinity levels, carbon limitation likely reduced photosynthetic capacity, reflected in lower Chl *a* concentrations. The consistency of the $\Delta TA_{\text{measured}} \geq \sim 1000 \mu\text{mol kg}^{-1}$ threshold across both dilution treatments during this short-term phase suggests that these high alkalinity (and low $f\text{CO}_2$) levels represent a fundamental physiological limit where carbon availability becomes critically constraining for the phytoplankton community. Notably, at $\Delta TA_{\text{measured}} \leq \sim 750 \mu\text{mol kg}^{-1}$, $f\text{CO}_2$ dropped as low as $37 \mu\text{atm}$ ($\text{pH} \leq 8.9$; referring to $\Delta TA_{\text{target}} 750 \mu\text{mol kg}^{-1}$) without a detectable effect on bulk Chl *a* concentrations (Fig. 5d). This suggests community-level resilience at low to moderate alkalinity levels even though $f\text{CO}_2$ concentrations reached here have been shown to impair growth in specific phytoplankton species in previous studies^{23,46–48}. While species-specific effects under carbon limitation cannot be excluded, the absence of a bulk Chl *a* signal at $\Delta TA_{\text{measured}} \leq \sim 750 \mu\text{mol kg}^{-1}$ indicates that any such effects were insufficient to alter community level Chl *a* within the 48-h exposure window. The 48-h period between alkalinity addition and mixing in the delayed dilution treatment is an ecologically relevant window for assessing short-term phytoplankton responses. In temperate eutrophic systems, typical cell doubling times for fast-growing phytoplankton such as diatoms can range from 12 to 48 h^{49–51}, allowing both physiological acclimation and early biomass responses to be detected. Moreover, Chl *a* concentrations integrate changes in growth and physiological state. These results demonstrate that OAE-induced carbonate chemistry perturbations can suppress phytoplankton biomass within ecologically relevant timescales at $\Delta TA_{\text{measured}} \geq \sim 1000 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \leq 23 \mu\text{atm}$ and $\text{pH} \geq 9.1$, Fig. 5c–f), when mixing of highly alkalinised waters is delayed. Given that large-scale OAE deployment will involve alkalinity additions to the surface ocean and mixing dynamics are location dependent, mapping these thresholds and rapid responses is critical for evaluating ecological risks and optimising delivery strategies to minimise unintended impacts.

OAE impacts on phytoplankton bloom phenology

The extent to which plankton communities can recover from acute stress following dilution of alkalinised waters to lower alkalinity levels is a critical question for assessing whether short-term, high intensity OAE exposures result in persistent ecological disruption or represent recoverable stress events.

Following the mixing of the top and bottom layers in the delayed dilution treatment, phytoplankton bloom dynamics were strongly modulated by alkalinity intensity. Irrespective of the dilution treatment, $\Delta TA_{\text{measured}} \leq \sim 700 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \geq 41 \mu\text{atm}$; values refer to $\Delta TA_{\text{measured}}$ levels on D7) showed minimal effects on bloom development compared to the controls, while $\Delta TA_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \leq 29 \mu\text{atm}$) delayed the day of maximum exponential growth by ~ 6 to 9 days (Fig. 7 and Supplementary Table 2). This reveals a critical threshold between $\Delta TA_{\text{measured}} \sim 700 \mu\text{mol kg}^{-1}$ (the $\Delta TA_{\text{measured}}$ showing no significant delay in bloom development) and $\sim 900 \mu\text{mol kg}^{-1}$ (the $\Delta TA_{\text{measured}}$ where significant bloom delay was observed), which is consistent across both dilution treatments as well as the short- and medium-term response phases, lending confidence to its robustness as an ecologically relevant boundary.

The timing of the bloom peak was not different between the dilution treatments, suggesting a recovery potential of the planktonic community

even following short-term exposure to extreme alkalinity ($\Delta TA_{\text{measured}} \geq 1600 \mu\text{mol kg}^{-1}$). In the delayed dilution treatments, where the top layer reached $\Delta TA_{\text{measured}}$ values of 1109 and $1197 \mu\text{mol kg}^{-1}$ before being reduced to $\Delta TA_{\text{measured}}$ 507 and $702 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2$ of 67 and $45 \mu\text{atm}$) after mixing, bloom development timing (maximum exponential growth) was ultimately unaffected (Fig. 7), despite initially suppressed Chl *a* concentrations in the top layer before dilution (Fig. 5c, d). We hypothesize that mixing with the bottom layer simultaneously introduced less carbonate chemistry-stressed phytoplankton and lowered alkalinity to tolerable levels below the identified threshold, enabling ecosystem recovery. In contrast, longer exposure to higher alkalinity and lower $f\text{CO}_2$ post-mixing ($\Delta TA_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$, $f\text{CO}_2 \leq 29 \mu\text{atm}$) resulted in persistent delays in bloom development across both dilution treatments. Notably, a recent study by Kousoulas et al.¹⁶ observed short-term growth delays in a Tasmanian summer phytoplankton community under non- CO_2 -equilibrated OAE at ΔTA of $500 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \sim 80 \mu\text{atm}$) but comparable bloom peak timing to controls, while Ferderer et al.³³ found no delays in a winter community from the same system despite $f\text{CO}_2$ levels as low as $\sim 70 \mu\text{atm}$. In contrast, Marín-Samper et al.⁵² reported minor alkalinity-induced delays in gross production, community composition and Chl *a* during a mesocosm experiment in post-spring bloom mesotrophic waters off Bergen, using non- CO_2 -equilibrated olivine and hydrated lime-inspired OAE with alkalinity additions of up to $600 \mu\text{mol kg}^{-1}$. Together, these findings highlight that phytoplankton responses to OAE are context- and threshold-dependent rather than exhibiting linear dose-responsive patterns with critical limits likely to be influenced by community composition, nutrient regime and the magnitude of alkalinity addition.

Neither macronutrient concentrations ($\text{Si}(\text{OH})_4$, PO_4 , NO_x) nor grazing pressure appear responsible for the observed bloom delays in our study, given that nutrients were not limiting immediately after mixing and zooplankton abundances were low and similar across treatments during the first half of the experiment⁵³, which is consistent with typical grazer dynamics during the phytoplankton spring bloom at the Helgoland Roads time series station⁴³. This points to carbonate chemistry perturbation as the primary driver of the delayed bloom development, rather than bottom-up or top-down controls.

The delayed phytoplankton bloom development observed under high alkalinity exposure likely reflects multiple interacting mechanisms within the phytoplankton community. Most prominently, extremely low $f\text{CO}_2$ levels in the highest ΔTA treatments (as low as $4 \mu\text{atm}$) likely imposed acute carbon limitation on the phytoplankton community, thereby suppressing phytoplankton growth^{23,45,54,55}. Shifts in carbonate speciation at high pH additionally alter the $\text{CO}_2/\text{HCO}_3^-$ balance, potentially favouring taxa with efficient CCMs, such as diatoms^{25,27,28}. While CCMs enable HCO_3^- utilisation under low CO_2 conditions, providing some phytoplankton groups a competitive advantage²⁸, upregulation of the necessary cellular machinery requires acclimation time and imposes energetic costs that could temporarily influence growth rates and nutrient uptake^{28,56,57}. These mechanisms could explain the initial Chl *a* suppression and delayed NO_x and $\text{Si}(\text{OH})_4$ uptake observed at $\Delta TA_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$, followed by gradual recovery as CCM-efficient taxa resume growth. Although our study did not measure CCM activity directly, the onset of bloom recovery following the delay in the day of maximum exponential growth under persistently stressful carbonate chemistry conditions at $\Delta TA_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$ (low $f\text{CO}_2$, high pH) provides robust evidence for biological acclimation through CCM upregulation. Our results align with known physiological responses expected under low CO_2 availability, where recovery depends on community composition and acclimation capacity^{28,57,58}.

Notably, the $f\text{CO}_2$ threshold associated with delays in the phytoplankton bloom development ($f\text{CO}_2 \leq 29 \mu\text{atm}$, $\text{pH} \geq 9$) was lower than previously shown to impact phytoplankton growth in laboratory studies^{23,46–48}. We hypothesize that the community-level resilience observed in this study emerges from physiological diversity, where CCM-efficient taxa can eventually compensate for initially stressed populations. Future studies should incorporate physiological or molecular markers of CCM

expression to directly link carbon acquisition strategies with phytoplankton community responses to OAE, enabling mechanistic predictions of ecosystem impacts across different phytoplankton assemblages.

In the highest ΔTA treatments ($\Delta TA_{\text{target}}$ of $1250 \mu\text{mol kg}^{-1}$), abiotic CaCO_3 precipitation caused simultaneous declines in TA and DIC, suggesting partial relief from stressful carbonate chemistry conditions and thus potential bloom recovery. However, $f\text{CO}_2$ remained largely unchanged during the precipitation period (Fig. 3b and Supplementary Table 1), indicating that precipitation-driven TA loss did not constitute a functionally relevant improvement in carbonate chemistry conditions for phytoplankton. Furthermore, Chl *a* concentrations in the immediate dilution treatment began increasing on D15, 8 days prior to the onset of precipitation on D23, demonstrating that bloom development preceded any precipitation-driven TA decline. This provides strong support that CCM upregulation and/or community succession towards CCM-efficient taxa, rather than chemical amelioration, were the primary mechanisms underlying phytoplankton bloom recovery under persistently stressful carbonate chemistry conditions. Nevertheless, we acknowledge that the coincidence of precipitation onset and boom recovery in the highest ΔTA delayed dilution treatment introduces a confounding observation that cannot be fully resolved with the current dataset.

The delays in bloom development identified here carry broader ecological implications through their potential to disrupt trophic interactions. One critical aspect of bloom timing is its role in structuring trophic interactions through the match/mismatch hypothesis⁵⁹, whereby temporal mismatches between prey availability and predator development stages can reduce recruitment success^{60,61}. On the other hand, delayed phytoplankton blooms may, in some cases, improve trophic alignment if they better coincide with grazer growth periods⁶². The ~6–9 days delay in bloom development observed at $\Delta TA_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$ could therefore alter trophic coupling during the spring bloom. With potentially significant consequences for zooplankton recruitment and higher trophic level production in OAE deployment regions. Whether such delays result in persistent ecological disruption or improve trophic alignment enough to compensate for trophic mismatches will depend on the degree to which grazers can track shifts in prey availability, a question that warrants dedicated investigation in future OAE ecological assessments.

The development of a secondary phytoplankton bloom in both dilution controls suggests that this could be an expected natural feature of this system under Si(OH)_4 and NO_x availability and unperturbed carbonate system conditions. The occurrence of a secondary bloom also in the lowest ΔTA level of the immediate dilution treatment ($\Delta TA_{\text{measured}} \sim 200 \mu\text{mol kg}^{-1}$) suggests minimal disruption of bloom patterns at this alkalinity level under well mixed conditions. In contrast, secondary blooms were absent in all delayed dilution treatments above $\Delta TA_{\text{measured}} 0 \mu\text{mol kg}^{-1}$, suggesting an even lower ΔTA threshold for bloom dynamics under delayed dilution scenarios. We hypothesize that this reflects the more intense carbonate chemistry perturbations experienced during the two-layer phase in these treatments, which may have altered phytoplankton community composition and microbial nutrient regeneration processes sufficiently to disrupt the succession sequence and nutrient basis necessary for secondary bloom formation.

The nutrient dynamics support a key role of Si(OH)_4 availability in secondary bloom dynamics: in mesocosms with a second bloom, Si(OH)_4 and NO_x remained available after the first bloom, while PO_4 was depleted. In contrast, in mesocosms without a second bloom, Si(OH)_4 and PO_4 were depleted simultaneously (Fig. 4 and Supplementary Fig. 12). These patterns suggest that the continued availability of Si(OH)_4 may have facilitated the development of a secondary, likely diatom-dominated bloom, as indicated by BSi concentrations (Supplementary Fig. 14), despite PO_4 limitation. Notably, in some mesocosms ($\Delta TA_{\text{measured}} \sim 450$ and $\sim 700 \mu\text{mol kg}^{-1}$ under immediate dilution), Si(OH)_4 dynamics suggest potential signs of a secondary bloom, although this pattern is only weakly reflected in Chl *a* concentrations. The absence of secondary blooms in treatments that completed a full first bloom cycle with sufficient time remaining for potential

secondary bloom development ($\Delta TA_{\text{measured}} \sim 200$ – $650 \mu\text{mol kg}^{-1}$ delayed; $\Delta TA_{\text{measured}} \sim 450$ – $700 \mu\text{mol kg}^{-1}$ immediate) implies an OAE effect on bloom succession rather than an artefact of the experimental duration.

For the highest ΔTA treatments ($\Delta TA_{\text{measured}} \geq 900 \mu\text{mol kg}^{-1}$), we cannot exclude that with the delayed bloom onset and prolonged Chl *a* specific bloom length, the primary and secondary bloom phases could have merged into a single extended bloom event. This consideration precludes definitive conclusions about secondary bloom suppression in the highest ΔTA treatments based on the current dataset. Grazing pressure is unlikely to have suppressed secondary bloom formation in high ΔTA treatments. Zooplankton abundances were negatively correlated with elevated alkalinity⁵³, reflecting relatively higher grazing pressure in the controls relative to alkalinised mesocosms. Overall, the combined results point towards the role of OAE in inducing changes in nutrient depletion patterns and phytoplankton succession that shape bloom re-occurrence.

OAE impacts on Chl *a* and particulate organic matter concentrations

The bloom delay observed in high ΔTA treatments raises a question of whether once the bloom recovered under persistently low $f\text{CO}_2$ conditions, the associated Chl *a* and POM dynamics were comparable to unaffected treatments, or whether the carbonate chemistry perturbation had lasting effects on Chl *a* and POM concentrations beyond the observed delay in bloom development. We assessed whether OAE affected the concentrations of Chl *a* and POM components during the key bloom development phase, defined as the interval from the maximum exponential growth to bloom termination.

Overall, our analysis revealed no pronounced TA/ $f\text{CO}_2$ -dependent impacts in mean POM concentrations across most POM components and OAE treatments. Across both dilution treatments, mesocosms treated with $\Delta TA_{\text{measured}} \leq 700 \mu\text{mol kg}^{-1}$ ($f\text{CO}_2 \geq 41 \mu\text{atm}$) showed no discernible shifts in bloom timing and average concentrations of Chl *a*, BSi, POC, PON and POP were comparable to controls (and the no-effect model), indicating robust maintenance of bloom development and biomass buildup. Furthermore, Chl *a* and BSi showed no significant OAE impacts/ $f\text{CO}_2$ -dependencies across all ΔTA and dilution treatments. Rather than reflecting an absence of a phytoplankton response, this pattern may also reflect the high physiological plasticity of these phytoplankton metrics, including short-term shifts in cellular Chl *a* content to maintain growth and photosynthesis under low $f\text{CO}_2$ ^{63–65}, changes in diatom silicification and community-level species replacements within phytoplankton communities, which could potentially mask OAE-driven responses^{66–69}. Notably, the absence of a detectable effect on BSi average concentrations during the key bloom development phase at any alkalinity level or in either dilution treatment contrasts with the analysis of Suessle et al.⁷⁰, who analysed data from the same mesocosm experiment. This discrepancy likely reflects methodological differences: Suessle et al.⁷⁰ defined the bloom phase based on Chl *a* alone and analysed the data with respect to pH as the focus of their study was abiotic BSi dissolution. In contrast, here we used $f\text{CO}_2$ as explanatory variable to assess biotic responses during bloom development and defined the bloom phase from nine independent estimates of the timing of maximum exponential growth and five estimates of bloom termination, resulting in an ecologically robust and comparable averaging window across treatments.

Beyond $\Delta TA_{\text{measured}} \sim 700 \mu\text{mol kg}^{-1}$, detectable responses emerged selectively with respect to POM components and in a dilution-dependent manner. POP emerged as the most sensitive biogeochemical indicator of OAE stress with lower average concentrations observed at $f\text{CO}_2 \leq 32 \mu\text{atm}$ (corresponding to $\Delta TA_{\text{measured}}$ levels $\geq 900 \mu\text{mol kg}^{-1}$ in this study) in both dilution treatments (Fig. 8 and Supplementary Fig. 20). Similarly, at these alkalinity and $f\text{CO}_2$ levels, PO_4 dynamics after mixing did not show a clear delayed decline in either dilution treatment, whereas NO_x and Si(OH)_4 showed delayed consumption patterns consistent with the phytoplankton dynamics observed in Chl *a* and BSi (Fig. 4 and Supplementary Figs. 13 and 14). This suggests that initial PO_4 uptake at $\Delta TA_{\text{measured}}$

$\geq \sim 900 \mu\text{mol kg}^{-1}$ may have occurred partly independently of growth and biomass buildup, which would be consistent with increased cellular phosphorus demand and luxury phosphorus uptake in excess of growth requirements under CO_2 limitation during the lag-phase of the bloom^{71–73}. It is well established that the upregulation of CCMs under CO_2 limitation requires the synthesis of phosphorus-rich RNA and enzymatic machinery, substantially increasing phosphorus demand in phytoplankton^{24,56,74}. As such, it has been shown that under low $f\text{CO}_2$, phytoplankton have a higher minimum phosphorus quota owing to CCM requirements⁷⁵. Simultaneously, the elevated ATP costs of CCM operation divert energy away from active phosphorus uptake and growth^{76,77}. Given the established interactions between CCM strategies and phosphorus dynamics⁷⁸, we hypothesize that the $f\text{CO}_2$ -dependent pattern in average POP concentrations observed across both dilution treatments potentially reflects a net effect of carbon acquisition, phosphorus uptake and allocation and growth at high alkalinities where $f\text{CO}_2$ is significantly limiting. We propose that if this supply-demand coupling drives phosphorus reallocation towards CCM support at the expense of growth and biomass buildup, internal phosphorus reserves could be depleted, suppressing POP accumulation in the water column⁷⁹. However, disentangling CCM-driven phosphorus reallocation from growth and biomass buildup would require direct measurements of cellular phosphorus quotas and CCM activity across $f\text{CO}_2$ gradients, and represents an important avenue for future OAE studies given our findings.

Significantly lower POC and PON concentrations were detectable only in the delayed dilution treatment at $f\text{CO}_2 \leq 32 \mu\text{atm}$ (corresponding to $\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$), with reductions of up to $\sim 11\%$ for POC and $\sim 21\%$ for PON relative to the no-effect model. This response appears to reflect a specific sensitivity of these biogeochemical parameters, as comparable dilution-related differences were not observed for other variables, including average Chl *a*, BSi and POP concentrations, as well as bloom timing. Moreover, similarly low $f\text{CO}_2$ levels were present in the immediate dilution treatment ($\leq 25 \mu\text{atm}$ vs. $\leq 29 \mu\text{atm}$ for the immediate and delayed dilution treatment on D7, respectively), yet mean POC and PON concentrations were not significantly lower under immediate dilution. This likely points to the role of prior exposure to stressful carbonate system conditions, rather than instantaneous $f\text{CO}_2$ alone, as a potential driver of lower average POC and PON concentrations. Under immediate dilution, the phytoplankton community in the two highest ΔTA treatments experienced lower alkalinity levels compared to the delayed dilution treatment during the first 48 h, which, despite delaying bloom development, ultimately allowed organic matter production to recover to control levels under blooming conditions. This is also consistent with higher peak POC and PON concentrations in the highest ΔTA immediate dilution mesocosms relative to their delayed counterparts (Supplementary Table 5), suggesting that the bloom dynamics were recoverable in terms of bulk organic matter concentrations. By contrast, in the delayed dilution treatment, 48 h exposure to very high alkalinity ($\Delta\text{TA}_{\text{measured}} > 1600 \mu\text{mol kg}^{-1}$, Fig. 5) and low $f\text{CO}_2$ (as low as $4 \mu\text{atm}$) during the two-layer phase appears to have imposed impacts that persisted beyond bloom recovery, resulting in a lasting and detectable reduction in mean POC and PON concentrations even after dilution to lower, but still elevated, ΔTA levels. Such effects of OAE on POC and PON concentrations could reflect OAE-induced changes in heterotrophic processes as well as shifts in diatom community composition³³, which might have shaped these dilution-dependent effects. Our findings are consistent with previous studies demonstrating that POC and PON concentrations can be influenced by carbonate chemistry speciation, though the magnitude of this effect may vary considerably depending on the composition of the plankton community^{13,33,80}.

Although CaCO_3 precipitation at the highest ΔTA level ($\Delta\text{TA}_{\text{target}} 1250 \mu\text{mol kg}^{-1}$) could in principle have enhanced particle ballasting and export, artificially depleting concentrations of suspended POM components⁸¹, two lines of evidence argue against it. First, similarly low POC and PON concentrations were not observed in the immediate

$\Delta\text{TA}_{\text{target}} 1250 \mu\text{mol kg}^{-1}$ treatment, which also precipitated. And second, Suessle et al.⁷⁰ conclude that CaCO_3 precipitation neither altered export magnitude nor increased particle sinking velocities.

Overall, our results clearly show that the estimated $f\text{CO}_2$ threshold range ($25\text{--}32 \mu\text{atm}$) was remarkably consistent across all POM components where a detectable response was identified and aligns with the threshold identified for bloom delay (Fig. 7), corresponding to $\Delta\text{TA}_{\text{measured}} \geq \sim 900 \mu\text{mol kg}^{-1}$. This convergence across various biogeochemical end-points (bloom timing and mean suspended POM biomass) supports the existence of a coherent OAE-influenced carbonate chemistry sensitivity threshold under spring bloom conditions, with the severity and duration of alkalinity exposure determining whether organic matter production is transiently delayed or lastingly impaired.

Biogeochemical feedbacks of ocean alkalinity enhancement: implications for organic matter cycling and stoichiometry

OAE altered the timing of the phytoplankton bloom and partly affected average concentrations of POM components during the key bloom development phase. However, these changes were not reflected in the POM stoichiometry. All elemental ratios (POC:PON, POC:POP and PON:POP) showed considerable variability across treatments and no consistent $f\text{CO}_2$ -dependent response over the full post-mixing period (D7 until D39) with all ratios remaining close to or slightly above Redfield proportions throughout most of the experiment (Fig. 9). The modest but consistent increase in POC:POP and PON:POP ratios above Redfield during the second half of the experiment is consistent with PO_4 being depleted early in most treatments while NO_x remained available throughout.

This observed stoichiometric resilience persisted across both dilution types and throughout the full experimental period despite OAE-induced shifts in bloom phenology, partly reduced average POM concentrations during the key bloom development phase and secondary bloom suppression. Thus, while OAE reduced the quantity of organic matter produced during the key bloom development phase, it did not detectably alter the quality of the bulk organic matter pool available to the food web over the full post-manipulation period, a distinction with important implications for assessing OAE impacts on trophic interactions and elemental cycling.

Bhaumik et al.⁸² showed that the nutritional quality of the cryptophyte *Rhodomonas salina*, a key zooplankton prey species, was maintained at $\Delta\text{TA} \leq 500 \mu\text{mol kg}^{-1}$. However, under elevated alkalinity above $\Delta\text{TA} 500 \mu\text{mol kg}^{-1}$, prey POC:PON and POC:POP ratios were lower (improved quality) despite significant reductions in prey growth. These species-specific improvement in phytoplankton nutritional quality (low POC:PON and POC:POP) observed by Bhaumik et al.⁸² was not apparent in our bulk community POM, where ratios were maintained rather than reduced relative to controls. This difference likely reflects community-level compensatory responses among phytoplankton taxa, with different stoichiometric sensitivities to OAE, as well as the mixed nature of bulk POM, which may mask species-specific changes. Together, these findings suggest that while individual phytoplankton species may alter their elemental composition under OAE stress, the bulk organic matter available to higher trophic levels maintains relatively stable nutritional quality across a wide range of OAE intensities, irrespective of dilution.

Conclusion

This study investigated short- and mid-term ecosystem responses to OAE, contrasting immediate and delayed dilution scenarios for OAE deployments. Overall, our results reveal a critical OAE threshold at $\Delta\text{TA}_{\text{measured}} \sim 700\text{--}900 \mu\text{mol kg}^{-1}$, corresponding to $f\text{CO}_2 \geq 41 \mu\text{atm}$, below which ecosystem resilience assessed in phytoplankton bloom timing, nutrient uptake, Chl *a* dynamics, as well as organic matter concentrations (POC, PON, POP and BSi) and stoichiometry remained largely unaffected. Above this threshold range, significant delays in bloom development of $\sim 6\text{--}9$ days and lower concentrations of specific POM components were observed. The observed threshold response is encouraging, indicating that low to moderate alkalinity additions of up to $\Delta\text{TA}_{\text{measured}} \sim 700 \mu\text{mol kg}^{-1}$

($f\text{CO}_2 \geq 41 \mu\text{atm}$) may be ecologically tolerable under spring bloom conditions in nutrient-rich, diatom-dominated systems (e.g., temperate eutrophic coastal waters). The phytoplankton community in the delayed dilution treatment demonstrated recovery potential from transient high alkalinity exposures when subsequent mixing after 48 h reduced alkalinity to lower levels below the identified threshold. However, across multiple ecological endpoints such as secondary bloom development, nutrient utilisation patterns and $f\text{CO}_2$ -dependent average concentrations of POM components, the delayed dilution treatment showed more pronounced and broader responses than the immediate dilution treatment at equivalent post-mixing alkalinity levels. The ecological safe operating space for OAE may therefore be more constrained under deployment scenarios with reduced mixing, reinforcing the importance of minimising prolonged exposure to strong carbonate chemistry conditions.

Defining a safe operating space for OAE will require comprehensive ecological assessments across the food web and seasonal regimes. Our findings provide a critical foundation for this effort by quantifying alkalinity and associated $f\text{CO}_2$ thresholds that distinguish resilient from disruptive ecosystem responses as well as demonstrating the pivotal role of dilution dynamics of OAE in modulating ecological impacts. Our findings, therefore, inform the development of deployment strategies and monitoring protocols that aim to balance carbon removal objectives with ecosystem preservation. As CDR advances from experimental to operational scales, such ecological boundaries will be essential for ensuring environmentally responsible implementation of OAE.

Material & methods

Study site and mesocosm deployment

The experiment was conducted on Helgoland, an island situated in the temperate-eutrophic waters of the German North Sea at the time series station Helgoland Roads (Dummermuth et al.⁸³; 54° 11' N, 7° 54' E). The experiment took place from March to April 2023. During this period, the plankton community is typically dominated by diatoms⁸⁴, experiencing low grazing pressure from zooplankton. Inorganic nutrient levels are generally elevated, ensuring that growth is not nutrient-limited, which, combined with low grazing, supports high microalgal growth rates⁴³.

On March 13th, twelve floating mesocosms were filled with seawater pumped from 2 to 3 m depth outside the harbour using a peristaltic pump (14 m³ h⁻¹, KUNZ SPF60, Flexodamp FD-50) and large-diameter hoses (Ø 37 mm). Each mesocosm was made of a transparent 1 mm thick polyurethane bag with a cylindrical upper section and a conical sediment trap at the bottom, extending to 3.5 m depth. The bags were mounted on floating frames and enclosed a seawater volume of ~6 m³. The mesocosms were filled simultaneously, ensuring that the collected water was distributed equally across all mesocosms. During the filling, the natural plankton community was gently transferred into the mesocosm bags, while larger organisms such as fish were excluded via pre-filtration with a 3 mm mesh. Mesocosms were covered with a lid that allowed for gas exchange while preventing the deposition of rain, large amounts of dust or other particles and were only opened during sampling and cleaning. The first sampling day (D1) of the experiment was on March 14th, with sampling continuing for 39 days, with final sampling on April 21st. Prior to the alkalinity treatment, a baseline sampling period was conducted over three days (D1–D3), allowing for establishing of sampling procedures and for the plankton community to recover from handling stress associated with the mesocosm filling process.

Experimental setup and TA manipulation

Alkalinity addition and water column mixing were conducted on D4 and D6, respectively, to simulate non- CO_2 -equilibrated OAE application under two dilution scenarios (Fig. 1). The alkaline chemicals (CaCl_2 and NaOH) were pre-dissolved in filtered freshwater immediately before addition, with freshwater serving as the carrier solution for alkalinity agents. Alkalinisation was achieved via the sequential addition of pre-dissolved calcium chloride (CaCl_2) and sodium hydroxide (NaOH), simulating the dissolution of calcium hydroxide (CaOH_2). For the immediate dilution scenario, designed to

mimic rapid dilution of alkaline material added to the surface seawater, six mesocosms were amended with total alkalinity addition (ΔTA) in increments of 250 $\mu\text{mol kg}^{-1}$ (target $\Delta\text{TA} = 0, 250, 500, 750, 1000, 1250 \mu\text{mol kg}^{-1}$). A second set of six mesocosms was used to simulate a delayed dilution scenario of alkalisated waters, mimicking point source alkalinity additions with slower mixing. For this, similar alkalinity levels as for the immediate treatment were added only to the upper 0.8 m of each mesocosm, aiming to achieve double the target ΔTA in there (target $\Delta\text{TA} = 0, 500, 1000, 1500, 2000, 2500 \mu\text{mol kg}^{-1}$). Mesocosms without added alkalinity ($\Delta\text{TA} 0 \mu\text{mol kg}^{-1}$) are referred to as “controls”.

To create stratification of top and bottom layers in the delayed dilution treatment, freshwater used as a carrier solution for alkalinity agents was added to the top 0.8 m of all six mesocosms. This stratification served to prevent mixing between the alkalisated top layers and un-alkalisated bottom layers over the 48-h two-layer phase. The total freshwater volume (51 L) represented <5% of the top layer water column volume, resulting in a salinity reduction of ~0.5 PSU in the top layer and ~0.3 PSU in the full water column, which is well within the natural salinity variability documented at the Helgoland Roads timeseries station⁴³. After 48 h (D6), the alkalisated layer was vertically mixed with the untreated bottom layer by bubbling air at high pressure from the sediment trap for one minute. Ultimately, this resulted in the same final ΔTA gradient as the immediate dilution treatment, but differed in the duration and intensity of plankton exposure to elevated alkalinity and pH. To ensure consistency among dilution treatments, an identical total freshwater volume of 51 L was also used as a carrier for alkalinity agents and added to the entire water column in the immediate dilution treatments on D4 (corresponding to ~1% of the total water column volume). In the controls, freshwater without alkalinity agents was also added in the same way, corresponding to the respective dilution manipulations. Additionally, air was bubbled through on D6 in all the immediate dilution treatments and controls as well. Further details on the manipulation are given in the Supplementary material.

Sampling procedures and maintenance

During the 39 experimental days, samples were taken every other day with the exception of days 5, 6 and 7, where daily sampling was conducted to monitor short-term responses after alkalinity manipulation (Supplementary Fig. 6).

Every sampling day, CTD casts were conducted first to ensure an undisturbed water column to provide vertical profiles of temperature, salinity, density, pH, turbidity, oxygen and photosynthetically active radiation (PAR) to 3 m depth. Next, depth-integrated water samples were collected using custom-made 2.5-meter-long plastic tubes with 5 L volume⁸⁵. Tubes with sample water were gently mixed before subsampling to ensure homogeneity across all parameters. To evaluate the stability of the induced stratification and the response to the different carbonate system conditions in top and bottom layers of the delayed dilution treatment, all parameters were sampled separately from the top and bottom layers on D5 and D6, using a Niskin sampler. Subsamples were collected in decreasing order of sensitivity to gas exchange, i.e., carbonate chemistry, nutrients and flow cytometry into respective collection containers. The rest of the water sample in the tube/Niskin sampler was collected into a 10 L carboy for later subsampling of other parameters (e.g., suspended particulate matter and photosynthetic pigments). At the end of each sampling, all samples were stored in a cool and sheltered location protected from direct sunlight.

To avoid wall growth on the mesocosm bags, mesocosms were scrubbed with brushes from the inside and outside six times and three times, respectively, during the experiment, exclusively on the non-sampling days (Supplementary Fig. 6).

Sample processing and measurements

Carbonate chemistry. For carbonate chemistry analysis, 500 ml of water was collected from the 5 L integrated sampling tube (or the Niskin sampler on D5 and D6) into sample rinsed glass stoppered 3.3

borosilicate glass bottles using a pre-rinsed Tygon tubing. To minimise gas exchange, samples were filled gently from the bottom to the top, ensuring an overflow of ~1.5 times the sample volume. Afterwards, samples were sterile filtered with a peristaltic pump through a polyether-sulfone filter (0.2 µm, Sartorius) to eliminate organic material and particulate inorganic carbon (PIC) particles. TA measurements were conducted in technical duplicates using a two-stage open-cell potentiometric titrator (titrant: 0.05 M HCl solution brought to an ionic strength of 0.72 M with NaCl; Compact Titrosampler 862, Metrohm, Munich, Germany), following the method outlined in Dickson et al.⁸⁶. All results were calibrated against certified reference materials (CRMs, batch 197), measured each sampling day and the instrument uncertainty was estimated at $\pm 2.0 \mu\text{mol kg}^{-1}$. TA was calculated using the titration curves and the “Calkulate” script within PyCO2SYS by Humphreys⁸⁷.

To measure dissolved inorganic carbon (DIC), sterile filtered triplicates were acidified with 10% phosphoric acid on an Automated Infra-Red Inorganic Carbon Analyzer (AIRICA, Marianda) to convert all DIC into CO₂. The CO₂ was then extracted using N₂ (5.0, ≥99.999% purity) and analysed with an IR detector LI-COR LI-7000⁸⁸. All measurements were calibrated against CRMs (batch 197) and the instrument uncertainty was estimated at $\pm 1.9 \mu\text{mol kg}^{-1}$.

Total scale pH and the fCO₂ were derived through calculations using the measured variables of DIC and TA, supplemented by temperature and salinity data derived from depth-integrated CTD casts. Calculations were executed using the MATLAB® version of CO2SYSv3⁸⁹, incorporating stoichiometric dissociation constants for carbonic acid from Sulpis et al.⁹⁰, for sulphuric acid from Dickson et al.⁸⁶ and total borate from Lee et al.⁹¹. Uncertainties were estimated to be ± 0.005 – 0.01 for pH and ± 1 – $11 \mu\text{atm}$ for fCO₂, with higher uncertainties at lower pH and higher fCO₂ concentrations, reflecting the nonlinear sensitivity of the carbonate system.

Chlorophyll a. For Chl *a*, subsamples (400–1000 ml) were collected from bulk water stored in 10 L carboys and gently filtered (< 200 mbar) onto 25 mm diameter glass fibre filters (GF/F, Whatman, 0.7 µm nominal pore size). The filtration racks were covered with aluminium foil during the filtration process to reduce light exposure. Filters were then transferred to 2 ml plastic vials (Sarstedt) and stored at -80°C for approximately 1.5 months prior to analysis at GEOMAR Kiel. Pigment extraction was performed using 100% acetone (Baker 8142, Avantor, Radnor, USA), followed by homogenization of the filters using 0.5 mm glass beads in a cell mill (Precellys, Montigny-le Bretonneux, France). After homogenization, samples were centrifuged (10 min, 10000 rpm, 4°C) and the resulting supernatant was filtered using a 0.2 µm polytetrafluoroethylene (PTFE) syringe filter (13 mm Ø, Lab Logistics Group). Concentrations of Chl *a* were quantified using reversed-phase high-performance liquid chromatography (Thermo Scientific HPLC Ultimate 3000), according to the protocol described by Van Heukelem & Thomas⁹².

Dissolved inorganic nutrients. For dissolved inorganic nutrients, 250 ml of water samples were collected from the sampling tube (or the Niskin sampler used for the top and bottom layer sampling on D5 and D6) into acid cleaned polypropylene bottles and stored in cool and dark conditions until analysis. The analysis of dissolved inorganic nutrients included the determination of nitrate (NO₃), nitrite (NO₂), (summed as dissolved inorganic nitrogen, NO_x), orthophosphate (PO₄) and silicic acid (Si(OH)₄). Samples were pre-filtered through a 0.45 µm PES-syringe filter (Sarstedt) to remove large particles before analysis. Subsequently, for each nutrient, triplicate samples were measured spectrophotometrically (Unicam UV 300, Thermo Spectronic, USA), following the methods outlined in H. Hansen & Koroleff⁹³. Values below the detection limit (LOD) were set to the corresponding LOD for each nutrient (PO₄ = $0.027 \mu\text{mol L}^{-1}$, Si(OH)₄ = $0.323 \mu\text{mol L}^{-1}$, NO_x = $0.467 \mu\text{mol L}^{-1}$).

Particulate organic matter. POM components (POC, PON and POP) represent bulk particulate pools integrating contributions from multiple

sources, including phytoplankton, heterotrophic organisms and detritus. To provide a more complete picture of the particulate matter composition, we report BSi, a phytoplankton-specific (particularly diatoms) indicator in the particulate matter pool alongside these POM components. Throughout, we collectively refer to POC, PON, POP and BSi as “POM components”, specifying individual parameters or distinguishing between phytoplankton-specific (BSi) and bulk POM pools (POC, PON, POP) where relevant.

Water samples for suspended POM were subsampled from 10 L carboys after gently mixing the carboys. Sample volumes ranged between 250 and 800 ml and were adjusted according to the bloom development to ensure enough material was collected without clogging the filters. All filtrations were performed using a vacuum filtration system (≤ 0.2 bar), within a filtration time of less than 30 min.

Water samples for POC and PON were filtered onto pre-combusted glass fibre filters (0.7 µm, GF/F Whatman, combustion at 450°C for 4 h) and acidified immediately after filtration (2 ml of 2 M HCl for 2 min) to remove particulate inorganic carbon (PIC). Subsequently, filters were collected in combusted glass Petri dishes (combustion at 450°C for 4 h) and dried at 60°C for 12 h before being packed into tin capsules and stored in desiccators for subsequent analysis at GEOMAR, Kiel, Germany. All samples were measured using an elemental analyser (Flash EA, ThermoFisher) according to Sharp⁹⁴. The CN content was calibrated using acetanilide (HEKAtech, catalog number HE33836700) and soil standard (HEKAtech, catalog number HE33860101) as standard reference materials.

For BSi determination, bulk water samples were filtered on cellulose acetate filters (0.7 µm, Sartorius Stedim Biotech GmbH), stored at -20°C and analysed within 4 days according to H. Hansen & Koroleff⁹³. Filters for BSi analysis were incubated in 25 ml 0.1 M NaOH at 85°C for 135 min to transform BSi to Si(OH)₄. Subsequently, dissolved silicic acid was measured spectrophotometrically following the methods outlined in H. Hansen & Koroleff⁹³.

Samples for POP were filtered onto pre-combusted glass fibre filters (0.7 µm, GF/F Whatman, combustion at 450°C for 4 h), collected into 2 ml microtubes (Eppendorf) and stored at -20°C until analysis at GEOMAR, Kiel, Germany. The filters were treated with 40 ml Milli-Q water and potassium peroxydisulphate (Oxisolv, MERCK) and autoclaved in a pressure cooker for 30 min to transform POP to dissolved PO₄. Dissolved PO₄ concentrations were analysed according to H. Hansen & Koroleff⁹³.

Data analysis

All data was quality controlled to identify and remove outliers before any further analysis. LODs were determined for each parameter, defined as the blank measurement variability $\mu \pm 3\sigma$. The blank consisted of 0.2 µm filtered seawater, processed in the same manner as the corresponding parameter analysis. R version 4.4.3. was utilized for data processing, statistical analysis and data visualization.

Acute responses to OAE (D5 and D6). Acute responses to alkalinity addition on D5 and D6 were assessed within each dilution treatment, focusing on treatment-specific effects. For the delayed dilution treatment, top and bottom layer samples were assessed to evaluate biological responses to the contrasting carbonate chemistry conditions along the ΔTA gradient, comparing the alkalinised top layer with the untreated bottom layer. For the immediate dilution treatment, which represents the intended well-mixed OAE scenario, responses were assessed along the fCO₂ gradient resulting from the ΔTA additions. Cross-dilution treatment comparisons between immediate and delayed dilution treatments during D5 and D6 were made through the fCO₂ framework relating biological responses to the carbonate chemistry conditions experienced in each sample type rather than as direct depth-matched comparisons. This approach is justified by our shallow mesocosm depth (3.5 m), resulting in uniform light and temperature conditions throughout the mesocosms' water column (Supplementary Figs. 1 and 2), which minimise depth-related physical confounding factors.

Post-mixing midterm responses. To capture potential delays in bloom development resulting from the alkalinity treatment, we focus on the development of the first bloom. For comparison purposes, two key bloom development stages were differentiated, which are unique to each individual mesocosm. In this way, two specific plankton ecosystem states could be used to contrast the differences between the treatments.

One key bloom development stage was defined as the day of maximum exponential growth within each mesocosm. To identify these points in time, we considered nine different types of observations (key biogeochemical parameters), using the measured concentrations of NO_3 , NO_2 , Si(OH)_4 , PO_4 , Chl *a*, BSi, POC, PON and POP. All data were first normalized to the smallest measured concentration to allow for comparability across variables. Rates of change were then calculated using a central difference approximation for an exponential growth model (Eq. 1). The central differences approximation reduced noise while smoothing short-term fluctuations in the data.

$$\mu = \frac{\ln(x(t + \Delta t)) - \ln(x(t - \Delta t))}{2\Delta t} \quad (1)$$

Here, $x(t)$ is the normalized concentration at time t . The rates of change μ are specific for the different types of observations and are all, since normalized, given in units 1/d. They correspond to relative growth rates (positive rates of change) of Chl *a*, BSi, POC, PON and POP, respectively. They represent relative nutrient utilization rates (negative rate of change; also in units of 1/d) when using the NO_3 , NO_2 , Si(OH)_4 and PO_4 data. The time interval Δt between the measurements is typically, although not exclusively, equal to two days.

For each mesocosm and variable, the day of the largest rate of change (either growth or uptake rate) was identified, yielding nine independent estimates of maximum growth per mesocosm. Because different processes respond at slightly different times, our approach to use multiple biogeochemical parameters provides a composite view of when the community collectively shifts from a stable to a growth-dominated state. Instead of estimating a single point in time for every mesocosm from a single type of measurements (e.g., Chl *a*), we first derived multiple estimates from all observational types. A best estimate of the time of maximum exponential growth is then derived from a probability distribution of these multiple estimates. For this purpose, we constructed nonparametric probability densities using kernel density estimation (KDE, Supplementary Fig. 9). Optimal bandwidths for the KDEs were determined using a quantile-based plug-in method⁹⁵. The mode of each probability density was used as the representative day of maximum exponential growth, allowing comparisons across ΔTA treatments at comparable plankton ecosystem states (Supplementary Table 2). Thus, each identified point in time is constrained by nine independent observational variables, resulting in a robust consensus estimate of the first key bloom development stage. The estimated times of maximum exponential growth for each mesocosm were subsequently used to quantify the temporal delay in bloom development (Fig. 7).

A similar approach was used to define a second key bloom development stage that is representative for the end of the bloom. In this case, the particulate matter concentrations were excluded. Only those data that best reflect different growth phases of the phytoplankton, i.e., dissolved inorganic nutrients (NO_3 , NO_2 , PO_4 , Si(OH)_4) and Chl *a* were included. Bloom termination is defined as the point in time at which the rate of change returned to zero, which corresponds to the cessation of net growth. According to this definition, five estimates of bloom termination per mesocosm could be obtained. As with the first key bloom development stage, KDEs were generated from these estimates, and again the mode of each probability density was then considered as a best estimate of the time of bloom termination (Supplementary Fig. 9 and Supplementary Table 2).

For the identification of changes in POM components induced by OAE, average concentrations of Chl *a*, BSi, POC, PON and POP are analysed individually. The analysis comprises three steps.

First, time-averaged concentrations of these five POM components (POM_i , $i = \text{Chl } a, \text{BSi}, \text{POC}, \text{PON}$ and POP) were calculated separately for each treatment (i.e., mesocosm). The averaging period spans the interval between the two key bloom development stages, the day of maximum exponential growth and the end of the bloom, which were identified individually for each mesocosm using KDE analysis (Fig. 9 and Supplementary Table 2).

Second, these averages were used to fit a nonlinear function describing the response to fCO_2 . Immediate and delayed dilution treatments were analysed separately. To describe potential effects of OAE on individual POM components, a single empirical model was employed, adapted from Equation 9 from Gafar et al.⁴⁴, with fCO_2 as the substrate and p_1 , p_2 and p_3 as model parameters (constant coefficients):

$$\text{POM}_i(\text{fCO}_2) = \frac{p_1 \text{fCO}_2}{p_2 \text{fCO}_2^2 + p_3 \text{fCO}_2 + 1} \quad (2)$$

Parameter estimation was performed by minimizing the root mean square error (RMSE), which served as the metric that quantifies the misfit between observed data and model results. For optimisation, the Covariance Matrix Adapting Evolutionary Strategy (CMA-ES) algorithm of N. Hansen⁹⁶ was applied, using the R package *cmaes* of Trautmann et al.⁹⁷.

In a third and final step, the robustness of the respective model fits was assessed relative to a no-effect model that is represented by the mean concentrations of the data across all ΔTA treatments. The mean concentration was determined for every POM component individually, and calculated independently for the respective dilution treatments. To identify potential effects induced by OAE on the different POM components, we defined an effect size in a bootstrap-based resampling framework. Specifically, 2000 samples were drawn with replacement from the observational data ($n = 6$ mesocosms per dilution type). These resampled datasets replicate the variability in the original data, thereby capturing variability across the ΔTA treatments. Every resampled dataset (of same size as number of observations, $n = 6$ mesocosms per dilution treatment) was used for CMA-ES optimization of the model's parameter values, yielding an ensemble of 2000 model solutions. When the data show high (random) variability across all treatments, then the ensemble of model fits may show a wide spread of plausible model solutions. If, however, the data show a systematic dependence on fCO_2 , the range of plausible model solutions becomes more constrained. Ultimately, for hypothesis testing, this ensemble of 2000 model solutions is then assessed using the (alternative) no-effect model as null hypothesis. For each ensemble member, deviations between the model solution and the no-effect model are determined (residuals). For the full ensemble of residuals, we can then determine whether the solutions of the fitted nonlinear model differ from the no-effect model. The effect size is defined as the change in residuals relative to the no-effect model. A detectable OAE response was inferred when the 10th–90th percentile range of the ensemble residuals did not include zero. The fCO_2 threshold value was derived as the fCO_2 level at which the 90th percentile of ensemble residuals crosses zero, indicating the onset of a detectable OAE effect for each POM component. The residuals and 10th–90th percentile range are shown in Supplementary Fig. 20, together with the derived fCO_2 threshold values.

The 10th–90th percentile range was used as a pragmatic measure of uncertainty that provides a balance between errors of Type I (falsely detecting an impact of OAE, although there is no effect) and Type II errors (missing the identification of a relevant impact). This choice reflects a more conservative approach to Type II errors, which prioritizes sensitivity to potential effects and reduces the probability of overlooking relevant OAE responses in a highly variable plankton ecosystem. At the same time, it limits the overinterpretation of stochastic variability, thereby reducing the risk of Type I errors. The selected percentile range thus represents a solid trade-off between sensitivity and specificity, capturing the prevailing variability while

also allowing for the identification of critical conditions under which an impact of OAE can be expected.

Data availability

The datasets presented in this study are openly available in the PANGAEA repository. Nutrient, chlorophyll *a* and particulate organic matter data are available at Tammen et al. (2026)⁹⁸ and carbonate chemistry data are available at Schulz et al. (2025)⁹⁹.

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Author contributions

J.K.T.: conceptualization, methodology, formal analysis (statistical modelling), investigation, visualization, writing—original draft, writing—review and editing; L.R.K.: conceptualization, methodology, investigation, supervision, writing—review and editing; M.S.: conceptualization, methodology, formal analysis (statistical modelling), visualization, supervision, writing—review and editing; G.F.: conceptualization, methodology, investigation, writing—review and editing; K.G.S.: conceptualization, methodology, supervision, writing—review and editing; M.B.: conceptualization, methodology, writing—review & editing; U.R.: conceptualization, methodology, supervision, writing—review and editing.

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Competing interests

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Additional information

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